COHERENT COHOMOLOGY OF SHIMURA VARIETIES, MOTIVIC COHOMOLOGY, AND ARCHIMEDEAN L-PACKETS

GYUJIN OH

Draft date: August 8, 2021

Abstract. We formulate an analogue of the archimedean motivic action conjecture of Prasanna–Venkatesh for *irregular* cohomological automorphic forms on Shimura varieties, which appear on multiple degrees of coherent cohomology of Shimura varieties. Such multiple appearances are due to many infinity types in a single L-packet with equal minimal K-types. Accordingly, we formulate the conjecture comparing periods of forms of *different automorphic representations*. We provide evidences for the conjecture by showing its compatibility with existing conjectures on periods of automorphic forms. In addition, guided by the conjectures, we suggest that there must be "*generalized complex conjugations*" (for the lack of a better name), certain geometric operations that can transform between different infinity types in an L-packet.

Contents

1. Introduction 2
1.1. Archimedean L-packet and the matter of choosing automorphic realizations 2
1.2. Comparison of periods of different automorphic representations 4
1.3. Summary 4
1.4. Problems and questions 4
1.5. Notation 5
2. Automorphic cohomology and archimedean L-packet 6
2.1. $(\mathfrak{p}, K)$-cohomology 6
2.2. Nondegenerate limit of discrete series as constituents of reducible principal series 7
2.3. Action of the Ext-space 10
2.4. Beilinson regulator 12
3. Formulation of the main conjectures 14
3.1. Metrics on cohomology 14
3.2. Archimedean motivic action conjecture for Shimura varieties 15
3.3. A concomitant conjecture on periods 15
3.4. Generalized complex conjugations 16
3.5. The motivic action conjecture on rationality of classes 18
4. Period integrals and coherent cohomology 19
4.1. The case of Sp_4 19
4.2. The case of $\mathrm{SU}(2,1)$ 21
5. Compatibility with conjectures on periods 21
5.1. Petersson norms as periods: doubling method and the GGP conjectures 22
5.2. Lapid–Mao conjecture 23
5.3. The case of Sp_4 23
5.4. The case of $\mathrm{SU}(2,1)$ 26
References 27

1. Introduction

The motivic action conjecture of Venkatesh posits that, roughly speaking, for a Hecke eigensystem h, there is a natural action of the motivic cohomology of the adjoint motive associated to h on the h-isotypic part of the rational cohomology of locally symmetric spaces. This has many incarnations which shed new light on various parts of the Langlands program. However, the conjectures have been mostly restricted to the case of "$\delta \neq 0$," namely when the reductive group in concern G has no compact Cartan subgroup. This in particular excludes the case when the locally symmetric space is a *Shimura variety*. As a locally symmetric space that is not a Shimura variety so far has not been related to algebraic geometry (although see [GT]), the motivic action conjectures seemed to be extremely difficult to approach.

On the other hand, there have been expectations that a similar conjecture would exist for automorphic forms over Shimura varieties with *irregular weight*. The easiest instance is the case of weight one modular forms; they appear in both H^0 and H^1 of the modular curve of the same line bundle. We give a formulation of such conjecture in a certain generality.

Conjecture 3.1 (Archimedean motivic action conjecture for Shimura varieties)**.** *Let λ be a nondegenerate singular analytically integral character, and let Π satisfy Assumption 2.1, with $\Pi_\infty \in \mathfrak{P}_\lambda$. Let $\mathcal{M}$ be the $\overline{\mathbb{Q}}$-vector space $H^1_M(\mathrm{Ad}\,\Pi, \overline{\mathbb{Q}}(1))$ equipped with a bilinear form which is the restriction of the hermitian metric on $H^1_{\mathscr{D}}(\mathrm{Ad}\,\Pi, \mathbb{C}(1))$ coming from the weak polarization of $\mathrm{Ad}\,\Pi$, as in* [PV, §2.2.4]*. Let $\mathcal{H}^i$ be the $\overline{\mathbb{Q}}$-vector space $H^i(X_G(\Gamma)_{\overline{\mathbb{Q}}}, [V_{A(\lambda)}])[\Pi_f]$ equipped with the restriction of the natural Hermitian metric. Then, there is an isomorphism of graded $\overline{\mathbb{Q}}$-vector spaces equipped with Hermitian metrics,*

$$\wedge^* \mathcal{M}^* \otimes \mathcal{H}^{i_l} \cong \bigoplus_{i=i_l}^{i_l+n_\lambda} \mathcal{H}^i.$$

This is an analogue of [PV, Conjecture 1.2.1]; indeed, the above conjecture seems quite similar. There are however quite a few differences and new phenomena that arise as we deal with irregular weights.

1.1. Archimedean L-packet and the matter of choosing automorphic realizations.

In view of automorphic cohomology, the phenomenon of a weight one modular form appearing in H^0 and H^1 actually involves *two different representations*. If we denote ω to be the weight one line bundle or the corresponding representation of $\mathrm{SO}(2)$, then for a weight one modular newform f,

$$H^0(X, \omega)[f] = H^0(\mathfrak{p}, \mathrm{SO}(2); D_0^+ \otimes \omega),$$
$$H^1(X, \omega)[f] = H^1(\mathfrak{p}, \mathrm{SO}(2); D_0^- \otimes \omega),$$

where $\mathfrak{p} = \mathfrak{p}_- \oplus \mathfrak{so}(2)$, $\mathfrak{p}_-$ is the anti-holomorphic tangent space and D_0^+ and D_0^- are the holomorphic and the antiholomorphic discrete series, respectively. Indeed, it is $\overline{f} d\overline{z}$ that appears in $H^1(X, \omega)$ as a class in Dolbeult cohomology, and $\overline{f}$ is an *antiholomorphic* modular form of weight -1.

In general, if the same Hecke eigensystem appears in multiple degrees of cohomology of the same automorphic vector bundle of a Shimura variety, then each such instance is actually represented by the so-called $(\mathfrak{p}, K)$-cohomology of different automorphic representations. More precisely, the finite part ($G(\mathbb{A}_f)$-representation) remains the same, while the infinity type varies

inside an archimedean L-packet. Such phenomenon happens precisely when the infinitesimal character of the infinity type lies on the walls of Weyl chambers.

The action of $\wedge^* \mathfrak{a}_G^*$ in [PV] is most naturally thought as the self-Ext-algebra of a representation. On the other hand, in our case, the target and the source of the action is different, so the action cannot be thought as an Ext-*algebra*, but merely as an Ext-group. Furthermore, the translation of the action into the automorphic cohomology context also depends on the choice of *automorphic realization maps* $\Pi_f \otimes \Pi_\infty \hookrightarrow \mathcal{A}(G)$ for each Π_∞; namely, each realization map can be always scaled by a scalar, but we want to compare them as a whole. This problem did not arise in *loc. cit.*, as a choice of a single automorphic realization map would rigidify the situation. In turn, we had no choice but to formulate a slightly weaker conjecture that asserts the motivic action conjecture on the level of metric spaces.

The main evidence of the conjecture lies on verifying a certain consequence of the conjecture on periods whose statement does not rely on motivic cohomology. The corresponding conjecture on periods is as follows.

Conjecture 3.2 (Period conjecture)**.** *Let $\Pi = \Pi_f \otimes \Pi_\infty$ be a cuspidal automorphic representation satisfying Assumption 2.1, and let λ be the infinitesimal character of Π_∞, which is singular and nondegenerate. Let V be the automorphic vector bundle coming from the Levi (see Notation), such that $H^i(\mathfrak{p}, K; V \otimes \Pi_\infty) \neq 0$ for some i, and $\Pi_{\min}, \Pi_{\max}$ be the members of the archimedean L-packet of Π_∞ such that the degree that $\Pi_{\min}$ ($\Pi_{\max}$, respectively) has nontrivial $(\mathfrak{p}, K)$-cohomology with coefficient in V is the minimum (maximum, respectively) in the L-packet. Pick $f_{\min} \in \Pi_f \otimes \Pi_{\min}$ and $f_{\max} \in \Pi_f \otimes \Pi_{\max}$, such that they define coherent cohomology classes in $H^*(X_G(\Gamma), V)$ which are defined over $\overline{\mathbb{Q}}$. Then,*

$$\frac{\langle f_{\min}, f_{\min} \rangle}{\langle f_{\max}, f_{\max} \rangle} \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} \left| \frac{L_\infty(1, \Pi, \mathrm{Ad})}{L_\infty(0, \Pi, \mathrm{Ad})} \cdot \frac{L(1, \Pi, \mathrm{Ad})}{\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\, \Pi)} \right|^2 .$$

At first sight, this might seem to be a less interesting conjecture, as it compares the top and bottom degree of cohomology. In [PV], the top and bottom degrees were complementary, so that they are related via duality. On the other hand, in the irregular setting, there are numerous cases where the top and bottom degrees are not complementary. Rather, they are determined by the position of the infinitesimal character inside the Weyl chambers. Indeed, we work with two examples, $G = \mathrm{Sp}_4$ and $\mathrm{SU}(2,1)$, and both cases we work with the choice of λ where the minimum and maximum degree of appearances are H^0 and H^1, respectively.

Towards formulating the archimedean motivic action conjecture in a full form determining rationality of cohomology classes, we realize that there should be a way that can rigidify between different automorphic representations in an L-packet. An example of this was complex conjugation for elliptic modular forms. We thus speculate the existence of *generalized complex conjugations*, which send an automorphic form of a certain infinity type to an automorphic form of another infinity type in the same L-packet. The existence of such action can be naturally understood in the context of a "derived" local-global compatiblity in some sense.

We expect that the operation of generalized complex conjugations should be constructed in a geometric way in some sense. Even though we do not have even a conjectural definition of what it should be, we investigate certain possibilities by poking with two constructions of geometric nature, *theta correspondence* and *(Carayol–)Penrose transform*. Both of these have a somewhat exotic effect on the infinity types, and by investigating how these interact with the usual complex conjugation, we speculate on what the generalized complex conjugations might be.

1.2. **Comparison of periods of different automorphic representations.** The main evidence for the Main Conjecture, Conjecture 3.1, is the verification of the Period Conjecture, Conjecture 3.2, after assuming several existing, well-accepted conjectures on periods. As does in [PV], we most notably assume *Beilinson's conjectures (for Chow motives).* The main new feature in this paper is that, because we need to compare periods of two different representations, we need two different conjectures on periods, one for each representation. In both evidences, we will compare periods of a *holomorphic automorphic form* and a *generic automorphic form.* For the holomorphic form, we will need the *refined Gan–Gross–Prasad conjecture* (referred as Ichino–Ikeda conjecture in *loc. cit.*), although in some instances this requirement can be avoided by using the doubling construction of standard L-functions. For the generic form, we will need the *Lapid–Mao conjecture*, which relates the value of a Whittaker function to a certain L-value.

One also needs a way to detect rationality of classes for both types of forms. For the holomorphic forms, we can use Fourier expansion, but for those appearing in higher coherent cohomology, we need a new machinery. We will develop the so-called *cohomological period integrals* for higher coherent cohomology, which realizes integral representations of L-functions as cup product pairings in coherent cohomology. The cohomological interpretation of such integral representations (or, at least, their appearance in the literature) is relatively new ([LPSZ], [Oh]).

1.3. **Summary.** In §2, we set up the language by using automorphic cohomology and representation theory of real reductive groups. In particular, we package several real group representations in our concern into a single reducible principal series. We then define the action of the Ext-group on the coherent cohomology of Shimura varieties, and relate the Ext-group with the motivic cohomology via Beilinson regulator, under the choice of automorphic realization map for each representation.

In §3, we formulate the main conjectures, most notably the Main Conjecture, Conjecture 3.1, and the Period Conjecture, Conjecture 3.2. We then investigate the possibility of the theory of *generalized complex conjugations* and state the conjecture on rationality of cohomology classes (Conjecture 3.3) using them. We also compare our conjectures with the existing work of [Ho] on archimedean motivic action conjecture on Hilbert modular forms, noting the discrepancy between it and our work.

In §4, we develop the machinery of cohomological period integrals for the groups of our concern, Sp_4 and $\mathrm{SU}(2,1)$. Using those apparatuses, we verify the compatibility of the Period Conjecture, Conjecture 3.2, with the existing conjectures on periods, such as Beilinson's conjectures, Lapid–Mao conjecture and refined Gan–Gross–Prasad conjectures.

1.4. **Problems and questions.** There are several interesting questions that arise in this work.

(1) Define (conjecturally) what the generalized complex conjugations should be. Place them in the context of some form of "derived" local-global compatibility.
(2) The compatibility between the Period Conjecture, Conjecture 3.2, and the existing period conjectures relies on the yet-to-be-calculated archimedean zeta integrals. These can be conducted using explicit integral formulae of (generalized) Whittaker functions, cf. [KO], [Od1].
(3) As we deal with motives over a more general number field and Shimura varieties over a number field other than $\mathbb{Q}$, in every aspect of our discussion, the choice of a complex embedding is always implicit. In particular, there must be a relation between the conjectures in this paper for the *conjugates of Shimura varieties* (cf. [Va]).

(4) It seems extremely hard to detect rationality of coherent cohomology classes if the Hecke operators can only cut a space that is of dimension larger than one. For example, if X is a Hilbert modular surface, and if ω is the parallel weight one line bundle, then it seems extremely difficult to determine whether a class in $H^1(X, \omega)$ is defined over $\overline{\mathbb{Q}}$, or even to produce a class in it.

1.5. **Notation.** Let G be a connected reductive algebraic group over $\mathbb{Q}$. For simplicity, let us assume that G is quasisplit, $G(\mathbb{R})$ is connected, and that the center of G does not have a nontrivial $\mathbb{R}$-split torus. Let $\mathfrak{g}_{\mathbb{Q}}$ be the $\mathbb{Q}$-Lie algebra of G, and let $\mathfrak{g}_{\mathbb{R}}, \mathfrak{g}_{\mathbb{C}}$ be its base change to $\mathbb{R}$ and $\mathbb{C}$, respectively. We occasionally drop the subscript for $\mathbb{C}$. Let W_G be the Weyl group of G. We endow an invariant, θ-invariant, $\mathbb{R}$-invariant bilinear form B on $\mathfrak{g}_{\mathbb{R}}$, such that $B(X, \theta(X))$ is negative definie, where θ is the Cartan involution. We will use this to talk about inner product on weight space, Riemannian metric on the Hermitian symmetric domain, etc. In specific examples, we may and will choose B to induce a preferred Riemannian metric on the Hermitian symmetric space (for example, one may want the Riemannian metric to be $dxdy$ on the upper half plane $\mathbb{H} = \{x + iy \mid y > 0\}$). We also use B to any other bilinear form induced from B.

Suppose further that G gives rise to a *Shimura variety*, which means that there is a symmetric space X for $G(\mathbb{R})$ which can be endowed with a structure of Hermitian symmetric domain (which we will fix). Fix a point $h \in X$, which gives rise to a *Hodge cocharacter* $h : \mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m,\mathbb{C}} \to G_{\mathbb{R}}$ which in turn induces a real Hodge structure of weight 0 on $\mathfrak{g}_{\mathbb{R}}$, $\mathfrak{g} = \mathfrak{g}^{-1,1} \oplus \mathfrak{g}^{0,0} \oplus \mathfrak{g}^{1,-1}$. Given an open compact subgroup $\Gamma \subset G(\mathbb{A}_f)$, there exists a quasi-projective variety $Y_G(\Gamma)$, a Shimura variety, defined over a number field E, whose complex points[1] have an analytification isomorphic to the double quotient $G(\mathbb{Q})\backslash(X \times G(\mathbb{A}_f)/\Gamma)$.

Let $K \subset G(\mathbb{R})$ be the stabilizer of h. Let T be the Cartan subgroup of K. Then, $X \cong G(\mathbb{R})/K$ and $\mathfrak{g}^{0,0} = \mathfrak{k} := \mathrm{Lie}(K)_{\mathbb{C}}$. We denote $\mathfrak{p}_+ = \mathfrak{g}^{-1,1}$, $\mathfrak{p}_- = \mathfrak{g}^{1,-1}$ and $\mathfrak{p} = \mathfrak{k} \oplus \mathfrak{p}_-$. Then, $\mathfrak{p}$ is a parabolic subalgebra of $\mathfrak{g}$, giving rise to a parabolic subgroup $P \subset G_{\mathbb{C}}$ with $\mathrm{Lie}\, P = \mathfrak{p}$. The holomorphic tangent space of X at h is identified with $\mathfrak{p}_+$, so there is a $G(\mathbb{R})$-equivariant embedding of complex manifolds $X \to \check{D} := G(\mathbb{C})/P(\mathbb{C})$, sending $h \mapsto P(\mathbb{C})$. In this regard, $K = G(\mathbb{R}) \cap P(\mathbb{C})$, and $K(\mathbb{C})$ is the Levi subgroup of $P(\mathbb{C})$. Also, any finite-dimensional holomorphic representation V of $P(\mathbb{C})$ gives rise to an algebraic vector bundle over $Y_G(\Gamma)$, an *automorphic vector bundle*, denoted $[V]$, which is an algebraization of the pullback of the holomorphic vector bundle on X which in turn is the restriction of the vector bundle $G(\mathbb{C}) \times^{P(\mathbb{C})} V \to \check{D}$ on $\check{D}$. If V factors through $K(\mathbb{C})$, namely if it is induced from a representation of $K(\mathbb{C})$, we will call $[V]$ an automorphic vector bundle *coming from the Levi*. If not, we will call V *nearly*, following [LPSZ].

To save space, we may abbreviate some words with repeated appearances: discrete series into DS, limit of discrete series into LDS, and nondegenerate limit of discrete series into NLDS.

For automorphic representations, their L-functions are normalized so that $\frac{1}{2}$ is the center of symmetry. For pure motives, theire L-functions are normalized so that, if w is its weight, $\frac{w+1}{2}$ is the center of symmetry. For both kinds of L-functions, w is called the *motivic weight* of the L-function.

[1]Note that the choice of a point in a Hermitian symmetric domain gives the reflex field as a subfield of $\mathbb{C}$, so there is a *preferred complex embedding*; cf. [Va, Notation 4.6]. In particular, one can expect that the statement of the conjecture depends a priori on the choice of a Hermitian symmetric domain. It could be interesting to check if our conjecture is consistent with conjugation of Shimura varieties.

For the integral representations, we fix a nontrivial additive character $\psi = \prod_p \psi_p$ of $\mathbb{A}/\mathbb{Q}$. For a cuspidal automorphic form $\varphi = \prod_p \varphi_p$ of cuspidal automorphic representation $\pi = \bigotimes_p' \pi_v$ of $G(\mathbb{A})$, its Whittaker transform $W_\varphi(g)$ is defined as $W_\varphi(g) = \int_{N_G(\mathbb{Q})\backslash N_G(\mathbb{A})} \varphi(ng)\psi(n^{-1})dn$, for some choice of N_G that needs to be specified when talking about Whittaker model. Locally, a generic $G(\mathbb{Q}_p)$-representation π_p is isomorphic to the space of functions $\mathcal{W}_p := \{W_p : G(\mathbb{Q}_p) \to \mathbb{C} \mid W_p(ng) = \psi_p(n)W_p(g) \text{ for } n \in N_G(\mathbb{Q}_p)\}$; one can choose isomorphisms (a local *Whittaker model*) $\pi_p \xrightarrow{\sim} \mathcal{W}_p$, $f_p \mapsto W_{f_p}$ that are compatible with the global Whittaker model; namely, $W_\varphi(g) = \prod_p W_{\varphi_p}(g_p)$.

We use notational convention for motives as in [PV, §2], except that we will deal with motives over a more general number field. In that case, we put the complex embedding in the subscript, such as M_σ, $\mathrm{comp}_{B,\mathrm{dR},\sigma}$, etc.

2. Automorphic cohomology and archimedean L-packet

2.1. $(\mathfrak{p}, K)$**-cohomology.** Recall that, as the singular cohomology of locally symmetric spaces can be calculated in terms of $(\mathfrak{g}, K)$-cohomology, the coherent cohomology of Shimura varieties can be calculated in terms of the so-called $(\mathfrak{p}, K)$-cohomology. By reinterpreting what the Dolbeault cohomology calculates in the setting of Shimura varieties, one gets the following

Proposition 2.1 (cf. [Su, (2.12)]). *We have*

$$H^i(Y_G(\Gamma), [V]) \cong H^i(\mathfrak{p}, K; C^\infty(G(\mathbb{Q})\backslash G(\mathbb{A})/\Gamma)^{K-\mathrm{finite}} \otimes V),$$

where V is understood as a $(\mathfrak{p}, K)$-module with trivial $\mathfrak{p}$-action.

Furthermore, there is an analogue of Franke's theorem for coherent cohomology of Shimura varieties.

Theorem 2.1 (Su, [Su, Theorem 6.7]). *For any sufficiently refined polyhedral cone decomposition Σ, there is a natural Hecke-equivariant isomorphism*

$$H^i(X_G^\Sigma(\Gamma), [V]^{\mathrm{can}}) \cong H^i(\mathfrak{p}, K; \mathcal{A}(G)^\Gamma \otimes V),$$

where $X_G^\Sigma(\Gamma)$ is the corresponding toroidal compactification, $[V]^{\mathrm{can}}$ is the canonical extension of $[V]$, and $\mathcal{A}(G)$ is the space of automorphic forms, namely the space of right K-finite, $Z(\mathfrak{g})$-finite smooth functions on $G(\mathbb{Q})\backslash G(\mathbb{A})$ of moderate growth.

As far as the coherent cohomology is concerned, the choice of Σ is ineffective, so we may occasionally drop the superscript Σ if there is no confusion.

The calculation of coherent cohomology is therefore about $(\mathfrak{p}, K)$-cohomology of automorphic representations. The $(\mathfrak{p}, K)$-cohomology of unitary $G(\mathbb{R})$-representations is computed by Vogan–Zuckerman. As far as $G(\mathbb{R})$-representations are concerned, we will be only interested in discrete series, or in general nondegenerate limits of discrete series. Recall that a nondegenerate limit of discrete series (NLDS) is a limit of discrete series whose infinitesimal character is not orthogonal to any compact root.

Theorem 2.2 (cf. [VZ], [Sc2]). *Let Π_∞ be the $(\mathfrak{g}, K)$-module associated to a DS or a NLDS representation of $G(\mathbb{R})$. Then, there is a unique $0 \le i \le \dim X$ and a finite-dimensional irreducible K-reperesentation V such that $H^i(\mathfrak{p}, K; \Pi_\infty \otimes V) \neq 0$. Furthermore, $\dim_{\mathbb{C}} H^i(\mathfrak{p}, K; \Pi_\infty \otimes V) = 1$.*

We will denote i_{Π_∞} and V_{Π_∞} for the i and V appearing in Theorem 2.2 corresponding to Π_∞.

2.2. **Nondegenerate limit of discrete series as constituents of reducible principal series.** We are interested in how the same Hecke eigensystem can appear in different degrees of coherent cohomology. As the Hecke eigensystem specifies only the *finite part* of automorphic representation, the analysis will involve $(\mathfrak{p}, K)$-cohomology of not just one $G(\mathbb{R})$-representation but rather *several representations in one archimedean L-packet.* For a (NL)DS, its L-packet consists of all (NL)DS with the same infinitesimal character. Thus each such L-packet is consisted of $|W_G|/|W_K|$ elements, and in particular, upon choosing a system of positive roots Δ_K^+ for K, can be indexed by Weyl chambers which makes all roots in Δ_K^+ positive. Given an infinitesimal character λ and a Weyl chamber C, let $A_C(\lambda)$ be the corresponding (L)DS. This is in accordance with the notation of [VZ]. In particular, $A_C(\lambda) = A_{\mathfrak{q}}(\lambda_C)$ for $\mathfrak{q} = \mathfrak{k} \oplus \mathfrak{p}_C$, where $\mathfrak{p}_C$ is the subalgebra of noncompact roots in Δ_C^+, the system of positive roots determined by C, and λ_C is the Weyl conjugate of λ that is contained in C. By the relatively straightforward nature of K-multiplicities of such representations, we have the following formulae.

Proposition 2.2. *Given λ and C as above, we have the following,*

$$i_{A_C(\lambda)} = \dim_{\mathbb{C}}(\mathfrak{p}_- \cap \mathfrak{p}_C),$$

$$V_{A_C(\lambda)} = V(\lambda_C + 2\rho(\mathfrak{p}_+)),$$

the highest weight representation of K with highest weight $\lambda_C + 2\rho(\mathfrak{p}_+)$, where $2\rho(\mathfrak{p}_+)$ is the sum of all roots in $\mathfrak{p}_+$.

Therefore, if λ lies on some wall of the Weyl chambers, then it can happen that $V_{A_C(\lambda)} = V_{A_{C'}(\lambda)}$ for different Weyl chambers C, C'. This is the setting we will be interested in. In such cases, we drop the subscript C if there is no issue of confusion.

Theorem 2.3. *Let λ be a nondegenerate analytically integral character. Let $\mathfrak{C}_\lambda = \{C \text{ Weyl chamber} \mid \lambda \in C\}$, and $\mathfrak{P}_\lambda = \{A_C(\lambda) \mid C \in \mathfrak{C}_\lambda\}$. Then, there is a parabolic subgroup $Q \subset G(\mathbb{R})$ and a discrete series representation ρ of the Levi M_Q such that*

$$I(\lambda) := \mathrm{Ind}_Q^G \rho = \bigoplus_{\pi \in \mathfrak{P}_\lambda} \pi,$$

where $\mathrm{Ind}_Q^G \rho$ is the normalized induction. These satisfy the following properties.

(1) Q is a parabolic subgroup which is minimal with respect to the property that the Langlands parameter $\varphi : W_{\mathbb{C}/\mathbb{R}} \to {}^L G$ corresponding to the representations in $\mathfrak{P}_\lambda$ can be arranged so that $\varphi(W_{\mathbb{C}/\mathbb{R}}) \subset {}^L Q$. Furthermore, if we denote T_{M_Q} by a Cartan subgroup of M_Q, then one can further assume that $\varphi(\mathbb{C}^\times) \subset {}^L T_{M_Q}$ and that $\varphi(W_{\mathbb{C}/\mathbb{R}})$ normalizes ${}^L T_{M_Q}$.
(2) $|\mathfrak{C}_\lambda|$ is a power of 2, and $\lambda^\perp \subset \mathfrak{g}$ is spanned by a superorthogonal set of real roots.
(3) The infinitesimal character of ρ is the restriction of λ.

Proof. We freely use the terminology of [Kn]. The property (3) is clear. The content of [Kn, §14.15] implies that any NLDS appears as a direct summand of such principal series with multiplicity one; this is via repeated application of generalized Schmid identities [Kn, Theorem 14.68]. The relation between Q and the Langlands parameter follows from the discussion before [La, Lemma 1]. Since the parabolic subgroup appears as Cayley transform of the minimal parabolic in the sense of [KZ1], and one chooses noncompact simple roots for the Cayley transform, which in

fact form a superorthogonal set of roots by [Kn, Theorem 14.64], based only on the infinitesimal character and not the chamber, it follows that

$$\{\mathrm{Ind}_Q^G \widetilde{\rho} \mid \widetilde{\rho}\,\mathrm{DS}, \text{infinitesimal character} = \text{the restriction of } \lambda\},$$

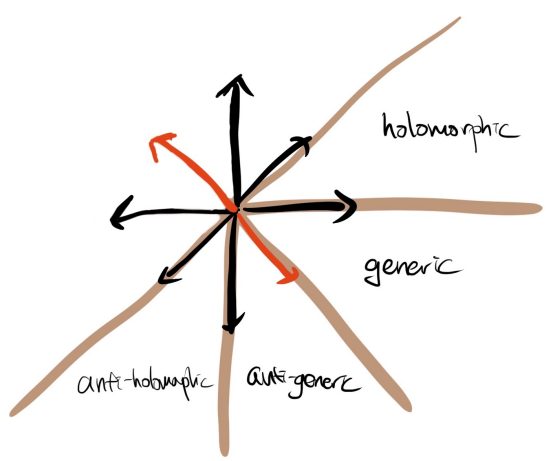
sees every NLDS with infinitesimal character λ at least once as its constituent. The number of constituents is exactly $|W_G|/|W_K|$, so each such NLDS appears exactly once. Each $\mathrm{Ind}_Q^G \rho$ has constituents $\pi = A_C(\lambda)$ such the representative of λ (thought as a Weyl orbit) in C is a fixed character (namely, C's that appear are all adjacent to a single representative of the Weyl orbit of λ). This is an equivalence relation, so this implies $I(\lambda)$ is precisely consisted of $A_C(\lambda)$ where C contains λ (regarded as an actual character). As the R-group is a direct sum of bunch of $\mathbb{Z}/2\mathbb{Z}$ by [Kn, §14.15], we get (2). $\square$

Example 2.1. We explain how Theorem 2.3 is realized in some examples. In the figures, red arrows are the compact roots, so NLDS's are those lying on a wall not orthogonal to red arrows.

(1) $\mathrm{SL}_2(\mathbb{R})$. Let $Q \subset \mathrm{SL}_2(\mathbb{R})$ be the upper triangular Borel, and let $\rho = \det/|\det| : Q \to \{\pm 1\}$. Then $I_Q^{\mathrm{SL}_2(\mathbb{R})}\rho = D_0^+ \oplus D_0^-$, the sum of the two NLDS, holomorphic ("weight 1") and anti-holomorphic ("weight -1").

(2) $\mathrm{Sp}_4(\mathbb{R})$ (cf. [Mu]). There are four types of LDS, two of them being holomorphic and anti-holomorphic, respectively, and the other two being large (i.e., maximal Gelfand–Kirillov dimension). We call the one adjacent to the holomorphic chamber *generic* and the other one adjacent to the anti-holomorphic chamber *anti-generic*. Using the notation of [Mu], a singular infinitesimal character is of one of the forms $(p, 0)$, $(0, -p)$ or $(p, -p)$ with $p \in \mathbb{N}$. The Langlands parameter φ_λ for infinitesimal character $\lambda = (a, b)$ is given by

$$\varphi_\lambda(re^{i\theta}) = \mathrm{diag}(e^{i(a+b)\theta}, e^{i(a-b)\theta}, e^{-i(a+b)\theta}, e^{-i(a-b)\theta}), \quad \varphi_\lambda(j) = \begin{pmatrix} 0 & (-1)^{a+b} I_2 \\ I_2 & 0 \end{pmatrix}.$$



- $\lambda = (p, 0)$ lies on the wall between the holomorphic chamber and the generic chamber. Then, Q is the so-called *Klingen parabolic*, whose Levi is $\mathrm{GL}_1(\mathbb{R}) \times \mathrm{SL}_2(\mathbb{R})$, and ρ is the (trivial extension of the) holomorphic DS D_p^+ of $\mathrm{SL}_2(\mathbb{R})$ of infinitesimal character p (or, equivalently, weight $p+1$).
- $\lambda = (p, -p)$ lies on the wall between the generic chamber and the anti-generic chamber. Then, Q is the so-called *Siegel parabolic*, whose Levi is $\mathrm{GL}_2(\mathbb{R})$, and ρ is the DS

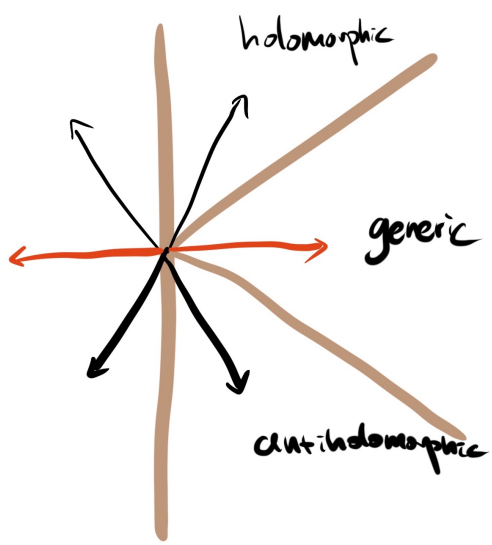
D_{2p} of $\mathrm{GL}_2(\mathbb{R})$ with central charcater $2p$ (which is as $\mathrm{SL}_2(\mathbb{R})$-representation the same as $D_{2p}^+ \oplus D_{2p}^-$).

- $\lambda = (0, -p)$ lies on the wall between the anti-generic chamber and the anti-holomorphic chamber. This situation is complex-conjugate to the situation of $\lambda = (p, 0)$. Thus, Q is again the Klingen parabolic, but ρ is the anti-holomorphic DS of the same infinitesimal character.

(3) $\mathrm{U}(2,1)$ (cf. [Wa], [Ro, §12]). There are three types of LDS, *holomorphic*, *generic* and *anti-holomorphic*. There are two typse of NLDS's , those lying on the wall between the holomorphic chamber and the generic chamber, and those lying on the wall betweent the generic chamber and the anti-holomorphic chamber. The Langlands parameters for (L)DS are of the form

$$\varphi(z) = \mathrm{diag}((z/\overline{z})^a, (z/\overline{z})^b, (z/\overline{z})^c), \quad \varphi(j) = \begin{pmatrix} & & 1 \\ & -1 & \\ 1 & & \end{pmatrix},$$

for $a, b, c \in \mathbb{Z}$, and the parameter only depends on the unordered set $\{a, b, c\}$. Ordering $a \geq b \geq c$, the two types of NLDS can occur for $a = b > c$ and $a > b = c$.



- $a = b > c$ lies on the wall between the holomorphic chamber and the generic chamber. Then, Q is the upper-triangular Borel (when $\mathrm{U}(2,1)$ is seen as the unitary group for the diagonal Hermitian matrix such as $\mathrm{diag}(1, 1, -1)$), whose Levi is $\mathbb{C}^\times \times S^1$, and ρ is the character $\mathrm{diag}(\alpha, \beta, \overline{\alpha}^{-1}) \mapsto \alpha^a \beta^b \overline{\alpha}^{-c}$ for $\alpha \in \mathbb{C}^\times$, $\beta \in S^1$.
- $a > b = c$ lies on the wall between the generic chamber and the anti-holomorphic chamber. The situation is completely symmetric to the previous case.

(4) $\mathrm{U}(2,2)$. There are six types of LDS, and those can be named as *holomorphic* (appears in H^0), *generic* (appears in H^1), *middle* and *anti-middle* (appears in H^2), *anti-generic* (appears in H^3), *anti-holomorphic* (appears in H^4). The Langlands parameters for (L)DS are of the

form

$$\varphi(z) = \mathrm{diag}((z/\overline{z})^a, (z/\overline{z})^b, (z/\overline{z})^c, (z/\overline{z})^d), \quad \varphi(j) = \begin{pmatrix} & & & 1 \\ & & 1 & \\ & 1 & & \\ 1 & & & \end{pmatrix},$$

for $a, b, c, d \in \mathbb{Z}$, and the parameter only depends on the unordered set $\{a, b, c, d\}$, so one can order $a \geq b \geq c \geq d$. We will not try to draw what happens due to the lack of drawing ability. We are interested in the case where the infinitesimal character lies on the wall between the holomorphic chamber and the generic chamber. This turns out to be the case when $a > b = c > d$; the generic chamber is obtained from the holomorphic chamber by reflection through the "$e_2 - e_3$" (in the usual $e_i - e_j$ notation for root system A_n), and this amounts to swapping the second largest number and the third largest number. In this case, Q is the parabolic subgroup of the form $\begin{pmatrix} * & * & * & * \\ & * & * & * \\ & * & * & * \\ & & & * \end{pmatrix}$.

2.3. **Action of the** Ext**-space.** We can now build an archimedean realization of motivic action from abstract nonsense. Recall that for a $(\mathfrak{p}, K)$-module M, $H^i(\mathfrak{p}, K; M)$ can be regarded as $\mathrm{Ext}^i_{(\mathfrak{p},K)}(\mathbf{1}, M)$, the i-th Ext group in the category of $(\mathfrak{p}, K)$-modules, where $\mathbf{1}$ is the trivial module. Thus, there is a natural action

$$\mathrm{Ext}^m_{(\mathfrak{p},K)}(M, N) \times \mathrm{Ext}^n_{(\mathfrak{p},K)}(\mathbf{1}, M) \to \mathrm{Ext}^{m+n}_{(\mathfrak{p},K)}(\mathbf{1}, N).$$

In particular, if λ is a nondegenerate singular character as above, with $\pi_1, \pi_2 \in \mathfrak{P}_\lambda$ with $i_{\pi_1} < i_{\pi_2}$, there is a natural action

$$\mathrm{Ext}^{i_{\pi_2} - i_{\pi_1}}_{(\mathfrak{p},K)}(\pi_1, \pi_2) \times H^{i_{\pi_1}}(\mathfrak{p}, K; \pi_1 \otimes V_{A(\lambda)}) \to H^{i_{\pi_2}}(\mathfrak{p}, K; \pi_2 \otimes V_{A_\lambda}),$$

where $V_{A(\lambda)}$ is the common coefficient for all $\pi \in \mathfrak{P}_\lambda$. Thanks to Theorem 2.3, the situation can be vastly simplified.

Theorem 2.4. *Let λ be as above.*

(1) There is a unique $\pi_l \in \mathfrak{P}_\lambda$ such that $i_{\pi_l} = i_l := \min\{i_\pi \mid \pi \in \mathfrak{P}_\lambda\}$.

(2) For each $\pi \in \mathfrak{P}_\lambda$, $\mathrm{Ext}^{i_\pi - i_{\pi_l}}_{(\mathfrak{p},K)}(\pi_l, \pi)$ is one-dimensional.

(3) Let $n_\lambda = \log_2 |\mathfrak{P}_\lambda|$, and $I_\lambda = \{\pi \in \mathfrak{P}_\lambda \mid i_\pi = i_{\pi_l} + 1\}$. Then, $n_\lambda = \#I$.

Proof. As in the proof of Theorem 2.3, [Kn, §14.15] gives us a recipe of how λ is constructed in terms of superorthogonal roots $\{\alpha_1, \cdots, \alpha_{n_\lambda}\}$, it follows that the chambers in $\mathfrak{C}_\lambda$ are of the following form: there is one chamber C_l that is the "most holomorphic" chamber (namely, $\dim(C_l \cap \mathfrak{p}^+)$ is the largest) and a set of superorthogonal noncompact roots $\{\alpha_1, \cdots, \alpha_n\} \subset C_l$ such that all other chambers $C \in \mathfrak{C}_\lambda$ are of the form

$$C_{\{i_1, \cdots, i_k\}} := (C_l - \{\alpha_{i_1}, \cdots, \alpha_{i_k}\}) \cup \{-\alpha_{i_l}, \cdots, -\alpha_{i_k}\}.$$

These imply (1) and (3). To show (2), we use the Hochschild–Serre spectral sequence for Lie algebra cohomology, [Vo1, Proposition 6.1.29]. In our setting, we apply a spectral sequence

$$E_2^{p,q} = \mathrm{Ext}^p_{(\mathfrak{k},K)}(X, H^q(\mathfrak{p}_-, Y)) \Rightarrow \mathrm{Ext}^{p+q}_{(\mathfrak{p},K)}(X, Y),$$

for $X = \pi_l$, $Y = \pi$. Since the category of $(\mathfrak{k}, K)$-modules is semisimple, there is no higher Ext and the spectral sequence trivially degnerates at the E_2-page:

$$\mathrm{Ext}^m_{(\mathfrak{p},K)}(\pi_l, \pi) = \mathrm{Hom}_{(\mathfrak{k},K)}(\pi_l, H^m(\mathfrak{p}_-, \pi)).$$

On the other hand, the main result of [Wi] says that, if we choose the maximal nilpotent subalgebra $\mathfrak{b}_- \subset \mathfrak{k}$ such that $\mathfrak{n} := \mathfrak{b}_- \oplus \mathfrak{p}_-$ is a maximal nilpotent subalgebra of $\mathfrak{g}$, then $H^*(\mathfrak{n}, \pi)$ can only have weights $W_K\lambda + \rho$, where ρ is the half sum of positive roots in $\mathfrak{g}$, where the positivity is defined so that $\mathfrak{n}$ is spanned by the negative roots. Furthermore, each such weight occurs in exactly one cohomological degree. To relate this result to our setting, we use another Hochschild–Serre spectral sequence,

$$E_2^{p,q} = H^p(\mathfrak{b}_-, H^q(\mathfrak{p}_-, \pi)) \Rightarrow H^{p+q}(\mathfrak{n}, \pi).$$

Suppose V_τ be a K-type that appears in $H^m(\mathfrak{p}_-, \pi)$, with highest weight τ. Then, by Kostant's theorem [Ko, Theorem 5.14], $H^p(\mathfrak{b}_-, V_\tau)$ has a nonzero μ-isotypic part if and only if $\mu = w(\tau + \rho_{\mathfrak{k}}) + \rho_{\mathfrak{k}}$ for some $w \in W_K$ of length p, where $\rho_{\mathfrak{k}}$ is the half sum of positive roots in $\mathfrak{k}$, where the positive system is defined so that $\mathfrak{b}_-$ is spanned by the negative roots in $\mathfrak{k}$. In particular, μ determines p and τ, which implies that this spectral sequence also degenerates at the E_2-page. This implies that the K-types of $H^*(\mathfrak{p}_-, \pi)$ are of highest weight $\lambda + \rho_{w\mathfrak{p}_-} - \rho_{\mathfrak{k}}$ for $w \in W_G/W_K$, and these are precisely the minimal K-types of $\pi' \in \mathfrak{P}_\lambda$.

Let τ_l be the highest weight of the minimal K-type of π_l. Then, all other minimal K-types of $\pi' \in \mathfrak{P}_\lambda$ are of highest weight of the form $\tau_l - \sum_i^{n_\lambda} \epsilon_i \alpha_i$ for $\epsilon_i \in \{0, 1\}$. By Blattner's formula for π_l [Sc2], all K-types of π_l are of highest weight $\tau_l + \delta$, where δ is a positive linear combination of noncompact positive simple roots, where positivity is defined by C_l. In particular, this is a cone lying in the direction towards $\alpha_1, \cdots, \alpha_{n_\lambda}$, so the only K-type that appears both in π_l and $H^m(\mathfrak{p}_-, \pi)$ are τ_l, whence $\dim \mathrm{Ext}^m_{(\mathfrak{p},K)}(\pi_l, \pi) \leq 1$. We have an exact formula for when nonzero Ext group occurs, which turns out to be $i_\pi - i_{\pi_l}$. Note that the same argument can be used to deduce the following: if $I \subset J \subset \{1, 2, \cdots, n_\lambda\}$, then $\mathrm{Ext}^{\#J-\#I}_{(\mathfrak{p},K)}(\pi_I, \pi_J)$ is one-dimensional. □

Proposition 2.3. *Upon choosing highest weight vectors of minimal K-types of $\pi \in \mathfrak{P}_\lambda$, for $I \subsetneq \{1, \cdots, n_\lambda\}$ with $i \notin I$, there is a natural identification*

$$\mathrm{Ext}^1_{(\mathfrak{p},K)}(\pi_I, \pi_{I\cup\{i\}}) \cong \mathbb{C}.$$

Proof. We think the Ext group as the group of actual extensions. In particular, $\mathrm{Ext}^1_{(\mathfrak{p},K)}(\pi_I, \pi_{I\cup\{\alpha_i\}})$ is naturally identified with the one-dimensional vector space of the short exact squences of $(\mathfrak{p}, K)$-modules,

$$0 \to \pi_{I\cup\{\alpha_i\}} \to V \to \pi_I \to 0.$$

Such extension is uniquely identified by a scalar $\lambda \in \mathbb{C}$ in a way as follows. The sequence splits as a short exact sequence of K-modules, so the minimal K-type τ_I of π_I has a canonical lift in V, also denoted as τ_I. Let $v \in \tau_I$ be the chosen highest weight vector. As $\alpha_i \notin I$, v as a vector in π_I satisfies $\alpha_i \cdot v = 0$, where α_i is seen as an element of $U(\mathfrak{p})$. However, if v is seen as a vector in V, $\alpha_i \cdot v$ is nonzero precisely when the extension is nonsplit, and is sent to the line of highest weight vector of the minimal K-type of $\pi_{I\cup\alpha_i\}}$. Upon fixing such a highest weight vector v', we have $\alpha_i \cdot v = \lambda v'$ for some $\lambda \in \mathbb{C}$, and this gives an isomorphism $\mathrm{Ext}^1_{(\mathfrak{p},K)}(\pi_I, \pi_{I\cup\{\alpha_i\}}) \cong \mathbb{C}$. □

Remark 2.1 (An important new difficulty in the $\delta = 0$ case)**.** It is very important to notice that, in our setting, the motivic action depends on the choice of *highest weight vectors of minimal K-types* of NLDS's, which was not necessary in the $\delta \neq 0$ case of [PV]. This is because our setting involves many different representations, whereas in the setting of [PV] one deals with a single representation. This will give rise to the conjecture on "generalized complex conjugations" in Section 3.4.

In the same vein, what naturally acts on the cohomology is not naturally an Ext-algebra, but merely the Ext-group $\mathrm{Ext}^1(I(\lambda), \pi_l)$. However, one can identify this with an exterior algebra as follows. There is a natural identification of $(\mathfrak{p}, K)$-cohomology of π_I for $I \subset \{1, \cdots, n_\lambda\}$,

$$\begin{aligned} H^*(\mathfrak{p}, K; \pi_I \otimes V_{A(\lambda)}) &= \mathrm{Hom}_K(\wedge^* \mathfrak{p}_-, \pi_I \otimes V_{A(\lambda)}) \\ &= \mathrm{Hom}_K(\wedge^* \mathfrak{p}_- \otimes \sigma_\emptyset, \sigma_I) \\ &= (\wedge^* \mathfrak{p}_+)^{\alpha_I}, \end{aligned}$$

where σ_J is the minimal K-type of π_J, and $\alpha_I = \sum_{i \in I} \alpha_i$. In particular, this is identified with $(\wedge^* \mathcal{E}_\lambda)^{\alpha_I}$, where $\mathcal{E}_\lambda \subset \mathfrak{p}_+$ is the span of $\{\alpha_1, \cdots, \alpha_{n_\lambda}\}$, because these form a superorthogonal set of roots. Thus, one can think of the exterior algebra $\bigwedge^* \mathcal{E}_\lambda$ as acting on the $(\mathfrak{p}, K)$-cohomologies of π's in $\mathfrak{P}_\lambda$. Note that, however, the choice of highest weight vectors of minimal K-types underlies everything.

We will also occasionally use another notation, $\lambda^\perp$, for $\mathcal{E}_\lambda$, to be more indicative of its definition.

The action can be connected back to coherent cohomology of Shimura varieties.

Assumption 2.1. Let $\Pi_f = \prod_{p<\infty} \Pi_p$ be an admissible $G(\mathbb{A}_f)$-representation, and λ be as above, namely a nondegenerate singular analytically integral character. We hereafter assume the following:

(MUL) For each $\Pi_\infty \in \mathfrak{P}_\lambda$, the automorphic multiplicity of $\Pi_f \otimes \Pi_\infty$ in $\mathcal{A}(G)$ is 1.

(NEW)
For each $p < \infty$, there exists a compact open subgroup $\Gamma_p \leq G(\mathbb{Z}_p)$ such that $\dim_{\mathbb{C}} \Pi_p^{\Gamma_p} = 1$.

In particular, when we say a vector $f \in \Pi$ is a *newform*, it means $f = \prod_p f_p$ such that, not only $f_p \in \Pi_p^{\Gamma_p}$, but also f_∞ is a highest weight vector of the minimal K-type of Π_∞.

Example 2.2. For example, upon assuming Arthur's conjectures, Assumption 2.1 is satisfied if Π_f is the finite part of a globally generic cuspidal automorphic representation $\Pi = \Pi_f \otimes \Pi_\infty$ of G, with Π_∞ being a generic LDS.

Under Assumption 2.1, let $\Gamma = \prod_p \Gamma_p$, and consider the Π_f-isotypic part of the coherent cohomology of $X_G(\Gamma)$ with coefficient $[V_{A(\lambda)}]$; by Theorem 2.1, we have a natural isomorphism

$$H^{i_{\Pi_\infty}}(X_G(\Gamma), [V_{A(\lambda)}])[\Pi_f] \cong H^i(\mathfrak{p}, K; \Pi_\infty \otimes V_{A(\lambda)}) \otimes_{\mathbb{C}} \Pi_f^\Gamma,$$

for each $\Pi_\infty \in \mathfrak{P}_\lambda$. Thus, using the notation of Theorem 2.4, there is an action

$$\wedge^* \mathcal{E}_\lambda \otimes_{\mathbb{C}} H^{i_l}(X_G(\Gamma), [V_{A(\lambda)}])[\Pi_f] \to H^{i_l + *}(X_G(\Gamma), [V_{A(\lambda)}])[\Pi_f].$$

2.4. **Beilinson regulator.** We now relate the Ext-space with motivic cohomology. This is to motivate the main conjecture, Conjecture 3.3; on the other hand, its formulation only needs what we have defined up to this point.

We first assume a conjecture on the existence of adjoint motive, similar to [PV, §4.2].

Conjecture 2.1. *For Π satisfying Assumption 2.1, there exists an adjoint motive $\mathrm{Ad}\,\Pi$, in the sense of* [PV, Definition 4.2.1]*, over the reflex field E with coefficients in F_Π, a totally real field.*

Note that one expects the Hecke eigensystem to be valued in a totally real field because of (NEW) of Assumption 2.1.

The Beilinson regulator for $\mathrm{Ad}\,\Pi$ is a map from motivic cohomology to Deligne cohomology,

$$H^1_M(\mathrm{Ad}\,\Pi, \mathbb{Q}(1)) \to H^1_{\mathscr{D}}((\mathrm{Ad}\,\Pi)_{\mathbb{C}}, \mathbb{R}(1)),$$

which conjecturally gives rise to an isomorphism $H^1_M(\mathrm{Ad}\,\Pi, \mathbb{Q}(1)) \otimes_{\mathbb{Q}} \mathbb{R} \xrightarrow{\sim} H^1_{\mathscr{D}}(\mathrm{Ad}\,\Pi, \mathbb{R}(1))$. By [PV, (2.2.8)] and the definition of adjoint motive in Conjecture 2.1, there is a natural isomorphism $H^1_{\mathscr{D}}((\mathrm{Ad}\,\Pi)_{\mathbb{C}}, \mathbb{R}(1)) \cong H_B((\mathrm{Ad}\,\Pi)_{\mathbb{C}}, \mathbb{R})^{W_{\mathbb{C}/\mathbb{R}}}$, and its complexification $H_B((\mathrm{Ad}\,\Pi)_{\mathbb{C}}, \mathbb{C})^{W_{\mathbb{C}/\mathbb{R}}}$ can be naturally identified with $\widehat{\mathfrak{g}}^{\varphi(W_{\mathbb{C}/\mathbb{R}})}$, the centralizer of the image of $W_{\mathbb{C}/\mathbb{R}}$ under the Archimedean Langlands parameter $\varphi : W_{\mathbb{C}/\mathbb{R}} \to {}^L G$.

Remark 2.2. Now that we are allowing motives over a more general number field other than $\mathbb{Q}$, we face a subtlety. Indeed, the Beilinson regulator is, conjecturally, the map induced from the Betti realization functor $\mathcal{MM}_E \to \mathcal{MH}_{\mathbb{R}}$ from the derived category of mixed motives over E to the derived category of mixed $\mathbb{R}$-Hodge structures, by realizing both H^1_M and $H^1_{\mathscr{D}}$ as some extension groups in $\mathcal{MM}$ and $\mathcal{MH}$, respectively. Note however that the Betti realization *depends on the complex embedding* $E \hookrightarrow \mathbb{C}$. In our discussion, we use the preferred embedding that came with the datum of reflex field. In the following discussions, we use this complex embedding unless we specify otherwise.

We arrange φ in a way that is done in Theorem 2.3. Then, the centralizer of $\varphi(\mathbb{C}^\times)$ is the Levi component $\widehat{M_Q A_Q}$, where $\mathrm{Lie}\,\widehat{M_Q A_Q} = \widehat{\mathfrak{t}} \oplus \widehat{\lambda^\perp_{\mathbb{C}}}$. Thus, $\mathrm{Lie}\,\widehat{M_Q}$ is a direct sum of n_λ-copies of $\mathfrak{sl}_2(\mathbb{C})$. Since the centralizer of $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ in $\mathfrak{sl}_2(\mathbb{C})$ is a one-dimensional torus, $\widehat{\mathfrak{g}}^{\varphi(W_{\mathbb{C}/\mathbb{R}})} = \mathbb{C}^{\oplus n_\lambda}$ is an n_λ-dimensional abelian Lie algebra, one coming from each α_i. It is therefore non-canonically isomorphic to $\lambda^\perp$. This also shows that $\widehat{\mathfrak{g}}^{\varphi(W_{\mathbb{C}/\mathbb{R}})}$ gives a Cartan subalgebra $\widehat{\mathfrak{h}} = \widehat{\mathfrak{t}} \cap \mathrm{Lie}\,\widehat{M_Q}$ of $\mathrm{Lie}\,\widehat{M_Q}$. Note that, similarly to [PV], the real structure for Deligne cohomology $H^1_{\mathscr{D}}(\mathrm{Ad}\,\Pi, \mathbb{R}(1)) \subset H^1_{\mathscr{D}}(\mathrm{Ad}\,\Pi, \mathbb{C}(1))$ transfers into the *twisted real structure* of $\widehat{\mathfrak{t}}$, as $\varphi(j)$ acts as the longest element of the Weyl group of $\widehat{M}$, so that the proof of [PV, Lemma 5.1.1] applies:

Definition 2.1. *The twisted real structure* $\widehat{\mathfrak{t}}'_{\mathbb{R}} \subset \widehat{\mathfrak{t}}$ *is defined by the involution* $(X \mapsto \overline{X}) \otimes w_0$*, where* $X \mapsto \overline{X}$ *is the involution coming from the usual real structure* $\widehat{\mathfrak{t}}_{\mathbb{R}} \subset \widehat{\mathfrak{t}}$*, and* w_0 *is the action of the longest element in the Weyl group of* $\widehat{M}$*.*

There is on the other hand a natural map $\widehat{\mathfrak{g}}^{\varphi(W_{\mathbb{C}/\mathbb{R}})} \to \mathrm{Ext}^1_{(\mathfrak{g},K)}(I(\lambda), \pi_h)$ whose definition does not depend on any arbitrary choice. Namely, twisting σ by a character of $H \subset M_Q$ gives a family of deformations of $I(\lambda) = \mathrm{Ind}_Q^G(\sigma)$, and the same deformation can be applied to extensions: namely, taking Ind_Q^G of $\mathrm{Ext}^*_{\mathfrak{h}}(\mathbf{1}, \mathbf{1})$ gives rise to a map

$$\mathrm{Ext}^*_{\mathfrak{h}}(\mathbf{1}, \mathbf{1}) \to \mathrm{Ext}^*_{(\mathfrak{g},K)}(\mathrm{Ind}_Q^G(\sigma), \mathrm{Ind}_Q^G(\sigma)).$$

Note that $\mathrm{Ext}^*_{\mathfrak{h}}(\mathbf{1}, \mathbf{1}) = \bigwedge^* \widehat{\mathfrak{h}}$, so we have a natural map

$$\wedge^* \widehat{\mathfrak{h}} = \mathrm{Ext}^*_{\mathfrak{h}}(\mathbf{1}, \mathbf{1}) \to \mathrm{Ext}^*_{(\mathfrak{g},K)}(I(\lambda), I(\lambda)) = \bigoplus_{\pi \in \mathfrak{P}_\lambda} \mathrm{Ext}^*_{(\mathfrak{g},K)}(I(\lambda), \pi) \twoheadrightarrow \mathrm{Ext}^*_{(\mathfrak{g},K)}(I(\lambda), \pi_h),$$

where the last map is the projection map. This isomorphism is natural, so this defines a *degree-descending* action of $\bigwedge^* H^1_{\mathscr{D}}(\mathrm{Ad}\,\Pi, \mathbb{C}(1))$ on $H^{i_h - *}(X_G(\Gamma), [V_{A(\lambda)}])[\Pi_f]$. Dually, this defines a *degree-ascending* action of $\bigwedge^* H^1_{\mathscr{D}}(\mathrm{Ad}\,\Pi, \mathbb{C}(1))^*$ on $H^{i_l + *}(X_G(\Gamma), [V_{A(\lambda)}])[\Pi_f]$.

Remark 2.3. Even though we have a definition of twisted real structure on $\widehat{\mathfrak{t}}$, this has no direct relationship with real structures on coherent cohomology of Shimura varieties. This is because

first of all the identification of the Ext-space with $\widehat{\mathfrak{t}}$ is not canonical. But more importantly, unlike the case of singular cohomology where $(\mathfrak{g}, K)$-cohomology is involved, $\mathfrak{p}$ is not invariant under the usual antiholomorphic involution. In particular, the coherent cohomology of Shimura varieties is defined over its reflex field E which is not necessarily a real field.

3. Formulation of the main conjectures

As we have emphasized several times above, the action of the Ext-group, and therefore the action of the Deligne cohomology, depends on the choice of highest weight vectors of the minimal K-types of $\pi \in \mathfrak{P}_\lambda$. Namely, one has to choose, for each $\pi \in \mathfrak{P}_\lambda$, an embedding $\Pi_f \otimes \pi \hookrightarrow \mathcal{A}(G)$, among the one-dimensional space of such embeddings (due to (MUL) of Assumption 2.1). Equivalently, one has to choose $f_\pi \in \mathcal{A}(G)$ for each $\pi \in \mathfrak{P}_\lambda$ such that it generates $\Pi_f \otimes \pi \hookrightarrow \mathcal{A}(G)$ and is

(A) a newform at each finite place (i.e. belongs to $\Pi_p^{\Gamma_p}$, as in (NEW) of Assumption 2.1),
(B) and a highest weight vector of the minimal K-type at archimedean place.

In this section, we will first state a weaker conjecture (the "metric version") which does not need to further claiify what f_π's should be. Then, we will transform the metric version conjecture into a conjecture on periods which may be verified without reference to motivic cohomology at all. Finally, we will formulate more speculative conjectures on both how to choose such vectors and the actual motivic action conjecture guided by the case of $\delta \neq 0$, and formulate the refined conjecture.

3.1. Metrics on cohomology.

A chosen bilinear form B on the Lie algebra naturally defines a Hermitian metric on the $(\mathfrak{p}, K)$-cohomology of unitary $(\mathfrak{p}, K)$-modules, or more generally, Ext^i between unitary $(\mathfrak{p}, K)$-modules. This is because the whole Chevalley–Eilenberg complex can be metrized. Furthermore, any natural map arising from abstract nonsense in Lie algebra cohomology is isometric. Since any finite-dimensional $(\mathfrak{p}, K)$-module can be endowed with an *admissible inner product* in the sense of [BW, II.2.2], this remark applies to the map

$$\mathrm{Ext}^*_{(\mathfrak{p},K)}(\pi_l, I(\lambda)) \otimes_{\mathbb{C}} H^{i_l}(\mathfrak{p}, K; \pi_l \otimes V_{A(\lambda)}) \to H^{i_l+*}(\mathfrak{p}, K; I(\lambda) \otimes V_{A(\lambda)}),$$

or using the Hermitian metric on the Dolbeault cohomology,

$$\mathrm{Ext}^*_{(\mathfrak{p},K)}(\pi_l, I(\lambda)) \otimes_{\mathbb{C}} H^{i_l}(X_G(\Gamma), [V_{A(\lambda)}])[\Pi_f] \to H^{i_l+*}(X_G(\Gamma), [V_{A(\lambda)}])[\Pi_f].$$

For $\mathrm{Ext}^*_{(\mathfrak{p},K)}(\pi_l, I(\lambda)) \cong \mathcal{E}_\lambda$ to be isometric, we need the factors coming from inner product on $I(\lambda)$ and π_l to cancel out, which means that the *norms of the chosen highest weight vectors of minimal K-types of $\pi \in \mathfrak{P}_\lambda$ to be equal.* The inner product can be given as either the L^2-norm on $G(\mathbb{Q})/G(\mathbb{A})$ with respect to the Tamagawa measure or the measure coming from the Riemannian metric, as the norms are all scaled by the same scalar factor. The second norm is the same as the usual *Petersson norm*, which we will denote as $\langle , \rangle_P$.

The discussion in this section suggests that the vectors $\{f_\pi \in \mathcal{A}(G) \mid \pi \in \mathfrak{P}_\lambda\}$ should satisfy, at least, the following additional condition:

(C) the Petersson norms $\langle f_\pi, f_\pi \rangle_P$ are all equal.

3.2. **Archimedean motivic action conjecture for Shimura varieties.** The conditions (A), (B), (C) unfortunately do not fully rigidify the f_π's as much as we would want to formulate the rationality conjecture as in [PV, §5.4]; namely, the choice of f_{π_l} does not determine the other f_π's. On the other hand, what we described is sufficient to formulate the "metric version" of the conjecture which, albeit weaker than the full conjecture on rationality, does not depend on choices.

Conjecture 3.1 (Archimedean motivic action conjecture for Shimura varieties, metric version)**.** *Let λ be a nondegenerate singular analytically integral character, and let Π satisfy Assumption 2.1, with $\Pi_\infty \in \mathfrak{P}_\lambda$. Let $\mathcal{M}$ be the $\overline{\mathbb{Q}}$-vector space $H^1_M(\mathrm{Ad}\,\Pi, \overline{\mathbb{Q}}(1))$ equipped with a bilinear form which is the restriction of the hermitian metric on $H^1_{\mathscr{D}}(\mathrm{Ad}\,\Pi, \mathbb{C}(1))$ coming from the weak polarization of $\mathrm{Ad}\,\Pi$, as in* [PV, §2.2.4]. *Let $\mathcal{H}^i$ be the $\overline{\mathbb{Q}}$-vector space $H^i(X_G(\Gamma)_{\overline{\mathbb{Q}}}, [V_{A(\lambda)}])[\Pi_f]$ equipped with the restriction of the natural Hermitian metric. Then, there is an isomorphism of graded $\overline{\mathbb{Q}}$-vector spaces equipped with Hermitian metrics,*

$$\wedge^* \mathcal{M}^* \otimes \mathcal{H}^{i_l} \cong \bigoplus_{i=i_l}^{i_l+n_\lambda} \mathcal{H}^i.$$

It seems that, to descend the coefficient field from $\overline{\mathbb{Q}}$ to a number field, one may have to take a field larger than $E(\Pi_f)$, the compositum of E and the field of definition of Π_f (cf. [Cl]), even in the case of $\mathrm{SL}_2(\mathbb{Q})$; see [Ho, Corollary 4.6].

3.3. **A concomitant conjecture on periods.** Assuming Beilinson's conjectures for Chow motives [PV, Conjecture 2.1.1], one can derive from Conjecture 3.1 a conjecture on periods that does not refer to motivic cohomology. The key is to compare the top and the bottom degrees of the graded vector spaces and relate them with the volumes, which is independent of the choice of metric. One can prove this in the identical way as it is done in [PV, Lemma 2.2.2], except that we work with motives over a more general field, not necessarily real, so we need to work with complex vector spaces with Hermitian metric as opposed to real vector spaces with symmetric inner product. Indeed, the fundamental exact sequences in [PV, 2.1.5] arise as "taking real points" (in the sense of Beilinson) of the analogous fundamental exact sequences of motives over $\mathbb{C}$, so we also have an exactly analogous formulation of Beilinson conjetcures with complex vector spaces with Hermitian structure.

Conjecture 3.2. *Let λ, Π as in Conjceture 3.1. Pick $f_l \in \Pi_f \otimes \pi_l$ and $f_h \in \Pi_f \otimes \pi_h$ such that they define coherent cohomology classes in $H^*(X_G(\Gamma), [V_{A(\lambda)}])$ which are defined over $\overline{\mathbb{Q}}$. Then,*

$$\frac{\langle f_l, f_l\rangle_P}{\langle f_h, f_h\rangle_P} \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} \left| \frac{L_\infty(1, \Pi, \mathrm{Ad})}{L_\infty(0, \Pi, \mathrm{Ad})} \cdot \frac{L(1, \Pi, \mathrm{Ad})}{\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\,\Pi)} \right|^2.$$

Example 3.1 (Sanity check: SL_2)**.** The simplest example is the case of $G = \mathrm{SL}_{2,\mathbb{Q}}$, where the conjecture is about *weight one elliptic modular forms.* Let $f \in S_1(\Gamma)$ be a weight one cuspidal new eigenform with Fourier coefficients in $\overline{\mathbb{Q}}$, generating an automorphic representation Π. Then, the complex conjugate $\overline{f} \in \overline{\Pi}$, which also satisfies $\langle \overline{f}, \overline{f}\rangle_P = \langle f, f\rangle_P$. Thus, we are led to the following question: for what $c \in \mathbb{C}^\times$ does $c\overline{f}$ define a $\overline{\mathbb{Q}}$-coherent cohomology class in $H^1(X(\Gamma), \omega)$? After calculating the archimedean L-factors (which is elementary), Conjecture 3.2 says that

$$|c|^{-2} \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} \left(\pi^{-2} \frac{L(1, \Pi, \mathrm{Ad})}{\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\,\Pi)} \right)^2.$$

Note that Π is a pure weight 0 motive, and so is $\operatorname{Ad}\Pi$; therefore, $F^1 H_{\mathrm{dR}}(\operatorname{Ad}\Pi) = 0$, which means that

$$|c| \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} \frac{\pi^2}{L(1,\Pi,\mathrm{Ad})}.$$

It is well-known that $L(1,\Pi,\mathrm{Ad})\pi^{-2} \sim_{\overline{\mathbb{Q}}^\times} \langle f, f\rangle_P$. Thus, the conjecture says

$$|c| \sim_{\overline{\mathbb{Q}}^\times \cap \mathbb{R}} \frac{1}{\langle f, f\rangle_P}.$$

On the other hand, it is also well-known that $\frac{\overline{f} d\overline{z}}{\langle f,f\rangle_P}$ defines a Dolbeault cohomology class in $H^1(X(\Gamma),\omega)$ which is defined over $\overline{\mathbb{Q}}$ as a coherent cohomology class. This means that $c = \frac{1}{\langle f,f\rangle_P}$, which is consistent with our conjecture. That these facts can be realized as an instance of motivic action conjecture was spelled out in [Ho].

Unlike [PV] or the case of SL_2, the top and bottom degrees are not complementary (namely, $i_l + i_h \neq \dim_{\mathbb{C}} X$), so the conjecture has nothing to do with any form of duality. We will nevertheless prove a form of the conjecture by relating this with *cohomological period integrals*.

3.4. **Generalized complex conjugations.** Example 3.1 suggests that the complex conjugation should play a role in the tentative statement of the full motivic action conjecture regarding rationality of cohomology classes. On the other hand, the complex conjugation can go back and forth between only two types of LDS's. For example, it sends a vector in the "holomorphic LDS" to that in the "anti-holomorphic LDS." Since a general motivic action conjecture involves many more LDS's, we suggest that there are "*generalized complex conjugations*" that can go between any of $\pi \in \mathfrak{P}_\lambda$. We denote a generalized complex conjugation, sending an automorphic form $v \in \Pi \otimes A_C(\lambda)$ to another automorphic form in $\Pi \otimes A_{C'}(\lambda)$, by $c_{C,C'}$. There are several desired properties:

- If $C_{\mathrm{hol}}, C_{\mathrm{antihol}}$ are in $\mathfrak{C}_\lambda$, $c_{C_{\mathrm{hol}},C_{\mathrm{antihol}}}(f) = \overline{f}$,
- $\langle f, f\rangle_P = \langle c_{C,C'}(f), c_{C,C'}(f)\rangle_P$ (Condition (C)),
- $c_{C',C''} \circ c_{C,C'} = c_{C,C''}$ and $c_{C,C} = \mathrm{id}$,
- $c_{C_1' \times C_2', C_1 \times C_2} = (c_{C_1',C_1}, c_{C_2',C_2})$, for $G(\mathbb{R}) = G_1(\mathbb{R}) \times G_2(\mathbb{R})$.

The last bullet point suggests that the *partial complex conjugation* (cf. [Ha2], [Sh2]) serves the role of generalized complex conjugations in the case of Hilbert modular forms; the partial complex conjugation is used in the conjecture of [Ho][2]. Unfortunately, aside from $\mathrm{SL}_2(\mathbb{R})$, it is almost never the case that C_{hol} and C_{anithol} are both in $\mathfrak{C}_\lambda$, because we do not allow degenerate λ; they are in some sense the two "furthest apart" chambers.

Although the desired properties are far from specifying what the generalized complex conjugations should be, we have a few suggestions on what they might be in certain nontrivial cases. The following two examples are some exotic constructions that are certainly "geometric", and if we believe that the generalized complex conjugation is a natural operation of geometric origin, then it can be believed that it should "intertwine" these other geometric constructions in some sense.

[2]Although we have not yet stated the rationality conjecture, the motivic action conjecture in [Ho] is not equivalent to our conjecture. We will comment about the discrepancy in more detail after we state our version of the conjecture.

Example 3.2 (Theta lifting)**.** In light of Vogan's reformulation of the local Langlands conjectures ([Vo2]), it has been realized that theta correspondences between a unitary group with itself preserves the L-packet but possibly changes the infinity type, in a way that the corresponding character of the component group is multiplied by a certain character ([Pr]). For example, if $\Pi = \Pi_f \otimes \Pi_\infty$ is a globally generic automorphic representation of $\mathrm{U}(2,1)$ with Π_∞ being the generic limit of discrete series with Harish–Chandra paraemter $(a, b; b)$ with $a > 0 > b$, then the theta lift of Π into the same group $\mathrm{U}(2,1)$ should be $\Pi_f \otimes \Pi_{\mathrm{hol}}$ where Π_{hol} is the holomorphic limit of discrete series with the same Harish–Chandra character.

Whether this is actually the desired generalized complex conjugation is arguable. However, this is a certain operation that interacts with the complex conjugation in a very nontrivial way. First, note that, in this case, the holomorphic chamber and the antiholomorphic chamber are never adjacent to each other, because this will mean that the Harish–Chandra character is zero, making the LDS's soleyl consisted of degenerate LDS. Rather, the complex conjugation sends Π to $\Pi_f \otimes \overline{\Pi}_\infty$, where $\overline{\Pi}_\infty$ is the generic LDS with Harish–Chandra parameter $(-b, -a; -b)$. This lies on the wall between the generic chamber and the anti-holomorphic chamber. Theta correspondence now sends this nontrivially to $\mathrm{U}(1,2)$ (the same group at the finite places, and the signature is flipped only at the archimedean place) to $\Pi_f \otimes \Pi'_\infty$, where Π'_∞ is the holomorphic LDS with Harish–Chandra parameter $(-b; -b, -a)$. Regarding this with the unique isomorphism $\mathrm{U}(1,2) \cong \mathrm{U}(2,1)$, its complex conjugation is the antiholomorphic LDS with Harish–Chandra parameter $(a, b; b)$. In short,

$$\begin{tikzcd}
\boxed{\text{Generic, } \lambda = (a,b;b)} \arrow[r, "\Theta"] \arrow[d, "c"'] & \boxed{\text{Holomorphic, } \lambda = (a,b;b)} \arrow[dr, "c"] & \\
\boxed{\text{Generic, } \lambda = (-b,-a;-b)} \arrow[dr, "\Theta"'] & & \boxed{\text{Antiholomorphic, } \lambda = (-b,-a;-b)} \arrow[dl, dashed, "L\text{-packet}"] \\
& \boxed{\text{Holomorphic, } \lambda = (-b,-a;-b)} &
\end{tikzcd}$$

If one thinks of theta correspondence as a geometric correspondence that should interact well with generalized complex conjugation, another "geometric correspondence", then one can speculate that the dotted arrow should be an instance of generalized complex conjugation.

A caveat here, however, is that the holomorphic LDS with Harish–Chandra parameter $(-b, -a; -b)$ is a *degenerate* LDS (thus is not an element in $\mathfrak{P}_\lambda$).

Example 3.3 (Carayol–Penrose transform)**.** There are mainly two ways of geometrically realizing (limits of) discrete series, one by solving certain PDEs on $G(\mathbb{R})/K$ coming from Dirac operators ([AS]) and one by using Dolbeault cohomology of various complex structures on $G(\mathbb{R})/T$ ([Sc1]). All types of (L)DS are equally dealt on $G(\mathbb{R})/T$, and Carayol (cf. [Ca]) used this idea to construct a geometric correspondence on the level of $G(\mathbb{R})/T \subset G(\mathbb{C})/B(\mathbb{C})$, which we will call *Carayol–Penrose transform*, and approached the *degenerate limit of discrete series* which is currently not detectable on the realm of cohomology of Hermitian symmetric spaces. The theory is more systematically developed in the context of flag domains in [GGK1], [GGK2], [Ke].

Recall that, under our assumption that $\mathrm{rank}_{\mathbb{R}} G = \mathrm{rank}_{\mathbb{R}} K$, there are $|W_G|/|W_K|$ many $G(\mathbb{R})$-orbits inside the flag variety $G(\mathbb{C})/B(\mathbb{C})$, and they are all homeomorphic to $G(\mathbb{R})/T$, so one can think of them giving $|W_G|/|W_K|$ many different complex structures on $G(\mathbb{R})/T$. Indeed, they

correspond to systems of positive roots with fixed subsystem of positive compact roots. We index them by Weyl chambers as we have been doing for representations, and let Z_C be the corresponding complex manifold. One can similarly define automorphic vector bundles for such domains. Then, the cohomology of Shimura varieties $X_G(\Gamma)$ can be read from the cohomology of $Z_{C_{\mathrm{hol}}}$, the flag domain corresponding to the holomorphic chamber; there is a holomorphic fibration $i : Z_{C_{\mathrm{hol}}} \to G(\mathbb{R})/K$, and $H^{i+a}(Y_G(\Gamma), V) \cong H^i(\Gamma\backslash Z_{C_{\mathrm{hol}}}, i^*V)$.

A basic feature of Carayol–Penrose transform that is relevant to our discussion is the following. It is a geometric correspondence between $Z_C, Z_{C'}$ such that it *shifts cohomological degree*: $H^i(\Gamma\backslash Z_C, V) \to H^i(\Gamma\backslash Z_{C'}, V')$, and it is often shown to be injective or even isomorphic. Here, the automorphic vector bundles V, V' are determined just by the minimal K-types. For example, in the case of Sp_4, it is shown that there is a Carayol–Penrose transform

$$CP : H^i(\Gamma\backslash Z_{C_{\mathrm{hol}}}, V) \to H^{i+1}(\Gamma\backslash Z_{C_{\mathrm{gen}}}, V'),$$

where C_{gen} is the "generic chamber" as in Example 2.1(2). Thus, via this transform, holomorphic forms are sent to H^1 whereas anti-generic forms are sent to H^3. Thus, again the usual complex conjugation interacts in an interesting way.

$$\begin{tikzcd}
\boxed{\text{Holomorphic, } H^0(\Gamma\backslash Z_{\mathrm{hol}}, V)} \arrow[r, dashed, "L\text{-packet}"] \arrow[d, "CP"'] & \boxed{\text{Generic, } H^1(\Gamma\backslash Z_{\mathrm{hol}}, V)} \arrow[r, dashed, bend left, "c"] & \boxed{\text{Antigeneric, } H^2(\Gamma\backslash Z_{\mathrm{hol}}, V^\vee)} \arrow[d, "CP"] \\
\boxed{H^1(\Gamma\backslash Z_{\mathrm{gen}}, V')} \arrow[rr, bend right, "c"'] & & \boxed{H^3(\Gamma\backslash Z_{\mathrm{gen}}, V'^\vee)}
\end{tikzcd}$$

Again, this realizes the Carayol–Penrose transform as a some sort of intertwiner between kinds of complex conjugations, and one may wonder if this gives rise to the desired generalized complex conjugation between holomorphic and generic forms (modulo showing that the map CP is an isomorphism).

3.5. **The motivic action conjecture on rationality of classes.** Under the hypothesis on existence of generalized complex conjugations $c_{C,C'}$, we can formulate the motivic action conjecture in the Shimura variety context in its full form.

Conjecture 3.3. *Let λ, Π as above. Let $f_l \in \Pi_f \otimes \pi_l$ be a newform that defines a $\overline{\mathbb{Q}}$-cohomology class. Under the hypothesis, choose the newforms for each $\{\Pi_f \otimes A_C(\lambda) \mid C \in \mathfrak{C}_\lambda\}$ to be $c_{C_l,C}(f_l) \in \Pi_f \otimes A_C(\lambda)$. This fixes the isomorphism $a : H^1_{\mathscr{D}}(\mathrm{Ad}^*\,\Pi, \mathbb{C}(1)) \xrightarrow{\sim} \mathrm{Ext}^1_{(\mathfrak{p},K)}(\pi_l, I(\lambda))$ as discussed above. Then, for $v \in H^1_M(\mathrm{Ad}^*\,\Pi, \overline{\mathbb{Q}}(1)) \subset H^1_{\mathscr{D}}(\mathrm{Ad}^*\,\Pi, \mathbb{C}(1))$, $a(v) \cdot f_l$ defines a coherent cohomology class $[a(v) \cdot f_l] \in H^{i_l+1}(X_G(\Gamma), [V_{A(\lambda)}])$ that is defined over $\overline{\mathbb{Q}}$.*

One can easily state a similar conjecture for the action of $\bigwedge^* H^1_M(\mathrm{Ad}\,\Pi, \overline{\mathbb{Q}}(1))$, but it is no deeper than the conjecture stated above.

There is basically one known case of what generalized complex conjugation should be, and it is the case of Hilbert modular forms.

Example 3.4 (The case of Hilbert modular forms; comparison with [Ho])**.** The work [Ho] states a similar conjecture, Conjecture 3.21 of *loc. cit.*, on what archimedean motivic action should be for Hilbert modular forms of partial weight one. The conjecture in particular implies that, for each $C \in \mathfrak{C}_\lambda$, a multiple of $f_C \in \Pi_f \otimes A_C(\lambda)$ defines a $\overline{\mathbb{Q}}$-class. This implies that the decomposition $H^i(X(\Gamma), \omega)[\Pi_f] = H^i(\mathfrak{p}, K; \omega \otimes I(\lambda)) \otimes \Pi_f^\Gamma = \left(\bigoplus_{\pi \in \mathfrak{P}_\lambda} H^i(\mathfrak{p}, K; \omega \otimes \pi)\right) \otimes \Pi_f^\Gamma$ descends to $\overline{\mathbb{Q}}$. There is no particular reason to believe this, because all the decomposition of cohomology of Shimura varieties defined over $\overline{\mathbb{Q}}$ come from Hecke correspondences (cf. [Ha1]), and in this case there is no Hecke operator to distinguish different infinity types.

The statement of conjecture 3.3 using some of the languages of [Ho] is as follows. Let λ define the discrete series corresponding to a partial weight one form. Each $C_{\{i\}} \in \mathfrak{C}_\lambda$, for $1 \leq i \leq n_\lambda$ correspond to a real embedding $\tau_i : F \to \mathbb{R}$. Then Conjecture 3.3 claims that

$$2\pi i \sum_{i=1}^{n_\lambda} \frac{\omega_f^{\{i\}}}{\log(|\tau \otimes \iota(u_{f,i})|)} \in H^1(X(\Gamma), \mathcal{E}),$$

defines a cohomology class over $\overline{\mathbb{Q}}$.

4. Period integrals and coherent cohomology

We will exhibit a set of evidences for Conjecture 3.2. Unlike [PV], we are to compare periods of two cohomological degrees which are not "dual" to each other. Thus, we are led to use a more sophisticated machinery, *cohomological period integrals*. These are integral representations of certain L-functions that

- apply to automorphic forms appearing in higher coherent cohomology,
- and admit an interpretation as cup product pairing in coherent cohomology.

This is useful in verifying our conjectures as the coherent cohomological cup product can *detect rationality of higher coherent cohomology classes*. In the following cases, we will compare periods of *holomorphic LDS* appearing in H^0 and *generic LDS* appearing H^1. The tools that we will use are summarized in the following table.

LDS type	Detecting rationality	Relation with Petersson norm
Holomorphic (H^0)	Rational Fourier coefficient	Doubling method, Refined GGP conjectures (§5.1)
Generic ($H^{>0}$)	Cohomological period integral (§4)	Lapid–Mao conjecture (§5.2)

Under these assumptions, and up to calculation of archimedean zeta integrals, we will show that our period conjecture, Conjecture 3.2, is true.

Theorem 4.1. *Let G be either Sp_4 or $\mathrm{SU}(2,2)$, and let Π be a globally generic cuspidal automorphic representation of $G(\mathbb{A}_\mathbb{Q})$ satisfying Assumption 2.1 and the assumptions of Conjecture 3.2. Assuming the working hypothesis on periods, Assumption 5.1, Conjecture 3.2 is true, up to the factor of an archimedean zeta integral.*

4.1. The case of Sp_4.

In this case, we will use an integral reprsentation of the spinor L-function by Novodvorsky ([No]), whose coherent cohomological interpretation was given by [LPSZ].

Let Π be a globally generic cuspidal automorphic representation of $G = \mathrm{GSp}_4(\mathbb{A}_\mathbb{Q})$, namely that the Whittaker transform (see §1.5, Notation) defines a realization of Π as a space of functions on $G(\mathbb{Q})\backslash G(\mathbb{A})$. We choose unramified vectors $\varphi_v^0 \in \Pi_v$ for v finite with Π_v unramified such that,

if ψ_v is unramified, $W_{\varphi_v^0}(1) = 1$. There is an embedding of $H = \mathrm{GL}_2 \times_{\mathrm{GL}_1} \mathrm{GL}_2$ into G, which is most naturally thought as $\mathrm{SO}(2,2) \hookrightarrow \mathrm{SO}(3,2)$. Let $B \subset H$ be the upper triangular Borel. For $\Phi_1, \Phi_2 : \mathbb{A}^2 \to \mathbb{C}$ and χ_1, χ_2 a unitary Grössencharcter, we define an Eisenstein series on H with respect to B, $E(h, \chi_i, \Phi_i, s_i)$. The following are well-known (cf. [LPSZ]).

Proposition 4.1 (cf. [Sh3]). *Suppose we are taking the "weight k-section" for $\Phi_{i,\infty}$.*

(1) For $-\frac{k}{2} + 1 \leq s_1, s_2 \leq \frac{k}{2}$ half-integral, equivalent modulo 1 to $\frac{k}{2}$, this defines a nearly holomorphic form on H, which can be thought as an H^0 class of some automorphic vector bundle over a Shimura variety for H.

(2) If χ_i's and Φ_i's are valued in a number field F, then the H^0 class is defined over F.

Assume now that Π satisfies Assumption 2.1, and let (λ_1, λ_2) be the Harish–Chandra character of the (necessarily generic) (L)DS Π_∞. Let $\Gamma_H = H \cap \Gamma$, and let $i : X_H(\Gamma_H) \hookrightarrow X_G(\Gamma)$ be the closed embedding of certain toroidal compactifications of the corresponding Shimura varieties[3]. One has the following cohomological interpretation of Novodvorsky's integral representation.

Theorem 4.2 ([LPSZ]). *Let $\varphi \in \Pi$ be a newform (see Assumption 2.1). Let $[\varphi] \in H^2(X_G(\Gamma), V)[\Pi_f]$ be the corresponding coherent cohomology class for the automorphic vector bundle V coming from the Levi. Let F be an imaginary quadratic field.*

Then, for any $s = \frac{w+1}{2} - i$, with $w = \lambda_1 - \lambda_2$ and $0 \leq i \leq w$ an integer, there is a nearly-holomorphic Eisenstein series E as above, corresponding to a certain choice of data, which defines a coherent cohomology class $[E] \in H^0(X_H(\Gamma_H), W)$ defined over FF_{Π_f}, such that the cup product pairing $\langle , \rangle$ defined by

$$H^2(X_G, V) \otimes_{\mathbb{C}} H^0(X_H, W) \xrightarrow{(\mathrm{id}, i_*)} H^2(X_G, V) \otimes_{\mathbb{C}} H^1(X_G, W') \xrightarrow{\cup} H^3(X_G, V \otimes W') \xrightarrow{S} H^0(X_G, \mathcal{O}) = \mathbb{C},$$

sends $([\varphi], [E])$ to a certain L-value,

$$\langle [\varphi], [E] \rangle = W_\varphi(1) \Lambda(s, \Pi) \Lambda(s, \Pi \otimes \chi_F),$$

where χ_F is the Hecke character corresponding to F, $\cup$ is the cohomological cup product, S is the Serre duality[4], normalized in a way that S is defined over $\mathbb{Q}$, and Λ is the completed L-function.

The key observation is that $\langle , \rangle$ is defined over $\mathbb{Q}$.

Corollary 4.1. *If $[\varphi]$ is defined over a number field $F' \supset FF_\Pi$, then*

$$W_\varphi(1) \sim_{F'^\times} \frac{1}{\Lambda(s, \Pi) \Lambda(s, \Pi \otimes \chi_F)},$$

where s is any half-integer as above with nonvanishing Λ-values.

For example, if one wishses, one can take the rightmost half-integer $s = \frac{w+1}{2}$. It then turns out that the Eisenstein series E should be taken to be a holomorphic (not just nearly holomorphic) Eisenstein series.

[3]That the closed immertion of open Shimura varieties extends to a closed immersion of toroidal compactifications with respect to certain refinements of polyhedral cone decomposition is achieved by [Lan], but the situation is simplier in this case, because X_H is unique.

[4]Thus, the Theorem implicitly contains $V \otimes W' \cong \Omega^3_{X_G}$.

4.2. **The case of** $\mathrm{SU}(2,1)$**.** In this case, we will use an integral reprsentation of the base-change L-function by Gelfand and Piatetski-Shapiro ([GPS], completed by [KO]), whose coherent cohomological interpretation was given by [Oh]. Let Π be a globally generic cuspidal automorphic representation of $G = \mathrm{SU}(2,1)(\mathbb{A}_{\mathbb{Q}})$ as in the previous subsection. There is an embedding of $H = \mathrm{U}(1,1)$ (what it really should be is $\mathrm{U}(1,1) \times_{\mathrm{U}(1,0)} \mathrm{U}(1,0)$) into G. Let $B \subset H$ be the upper-triangular Borel. For $\Phi : \mathbb{A}^2 \to \mathbb{C}$ and χ a unitary Grössencharacter of the imaginary quadratic field F used for the definition of unitary groups, we define an Eisenstein series on H with respect to B, $E(h, \chi, \Phi, s)$. The same arithmeticity condition as Proposition 4.1 applies; in both cases of concern, we are just dealing with elliptic modular forms on H.

Assume now that Π satisfies Assumption 2.1, and, using the notation of Example 2.1(3), let $(m,n) = (a-b, b-c)$ be the Harish–Chandra character of Π_∞ (note that G is now $\mathrm{SU}(2,1)$, so only the differences matter). Let $\Gamma_H = H \cap \Gamma$, and let $i : X_H(\Gamma_H) \hookrightarrow X_G(\Gamma)$ be the closed embedding of closed Shimura varieties, as above. One has the following cohomological interpretation of Gelfand–Piatetski-Shapiro's integral representation.

Theorem 4.3 ([GPS], [KO], [Oh])**.** *Let $\varphi \in \Pi$ be a newform (see Assumption 2.1). Let $[\varphi] \in H^1(X_G(\Gamma), V)[\Pi_f]$ be the corresponding coherent cohomology class for the automorphic vector bundle V coming from the Levi.*

Then, for any $s = \frac{w+1}{2} - i$, with $w = |m-n|$ and $0 \le i \le w$ an integer, there is a nearly-holomorphic Eisenstein series E as above, corresponding to a certain choice of data, which defines a coherent cohomology class $[E] \in H^0(X_H(\Gamma_H), W)$ defined over FF_{Π_f}, such that the cup product pairing $\langle , \rangle$ defined by

$$H^1(X_G, V) \otimes_{\mathbb{C}} H^0(X_H, W) \xrightarrow{(\mathrm{id}, i_*)} H^1(X_G, V) \otimes_{\mathbb{C}} H^1(X_G, W') \xrightarrow{\cup} H^2(X_G, V \otimes W') \xrightarrow{S} H^0(X_G, \mathcal{O}) = \mathbb{C},$$

sends $([\varphi], [E])$ to a certain L-value,

$$\langle [\varphi], [E] \rangle = W_\varphi(1) \Lambda\left(s, BC(\Pi)\right),$$

where BC means base-change, $\cup$ is the cohomological cup product, S is the Serre duality[5]*, normalized in a way that S is defined over $\mathbb{Q}$, and Λ is the completed L-function.*

Corollary 4.2. *If $[\varphi]$ is defined over a number field $F' \supset FF_\Pi$, then*

$$W_\varphi(1) \sim_{F'^\times} \frac{1}{\Lambda\left(s, BC(\Pi)\right)},$$

where s is any half-integer as above with nonvanishing Λ-values.

5. Compatibility with conjectures on periods

We will use the cohomological period pairings in the previous section to show one of our main results, Theorem 4.1, which is the primary evidence we have for Conjecture 3.2.

Assumption 5.1. We assume the following conjectures[6]. There are numerous instances of these conjectures being verified, and we do not attempt to list them here.

(1) *Beilinson's conjecture for Chow motives*, Conjecture 2.1.1 of [PV].

[5] Again, the Theorem implicitly contains $V \otimes W' \cong \Omega^2_{X_G}$.

[6] Even though the refined Gan–Gross–Prasad conjectures are being mentioned throughout the paper, they are not required as an assumption, because we have an unconditional alternative result using doubling method (and they are equivalent if one assumes Beilinson's conjectures anyways).

(2) *Lapid–Mao conjecture*, Conjecture 5.1.

The first main difference between our working hypothesis and that of [PV] is that we need an additional conjecture on periods. Here, we use the Lapid–Mao conjecture, as in both cases of our concern, the non-holomorphic automorphic form we will compare belongs to a generic (L)DS, which is a numerical coincidence that only happens in low-dimensional examples.

5.1. **Petersson norms as periods: doubling method and the GGP conjectures.** Starting from [Sh1], there have been many instances relating motivic periods and Petersson norm of holomorphic automorphic forms. As in the case of cohomological period pairing, this is proved by using some integral representation of an L-function. What's additionally useful about holomorphic automorphic forms is that its rationality as an algebraic section can be detected by the rationality of certain periods, such as Fourier(–Jacobi) coefficients. The simplest example of such is a criterion for algebraicity of elliptic modular form; a normalized cusp form is algebraic, namely the corresponding section is defined over $\overline{\mathbb{Q}}$, if and only if any nontrivial coefficient (or, equivalently, all of the coefficients) in its q-expansion is in $\overline{\mathbb{Q}}$.

On the other hand, the major drawback of holomorphic automorphic forms is that finding an integral representation for an L-function is more difficult, because it is hard to find a model that the representation admits. Apart from the case of Hilbert modular forms, holomorphic automorphic forms generate a representation that never admits a Whittaker model. In some sense, rather, holomorphic representations are the "most degenerate" (having the lowest Gelfand–Kirillov dimension, admitting the most degenerate Whittaker model). This problem is however brilliantly resolved by the *doubling method* of Piatetski-Shapiro and Rallis ([GPSR]) for the standard L-function, where the idea is to consider two copies of the same group and shove them into a larger group. In this way, the Petersson norm of a holomorphic automorphic form and its standard L-value are related in the case of classical groups ([Bö], [Ha3], [Ha4], [Sh3]). The end result is of the form

$$\text{Petersson norm of ratoinal holomorphic form} \sim \text{Standard } L\text{-value}.$$

On another note, one can start from a Fourier coefficient and track backwards. Note that the term "Fourier expansion" may mean many things, especially if one goes to higher dimensional groups. For example, for a Siegel modular form, there are two well-known Fourier expansions, namely q-expansions, and Fourier–Jacobi expansions. Conjecturally, each parabolic subgroup would give rise to a theory of Fourier expansions ([Od2]), with the q-expansion and the Fourier–Jacobi expansion being associated with the Siegel parabolic and the Jacobi parabolic, respectively. Each Fourier expansion is related to a certain period that is, roughly speaking, the integral over the parabolic against another cusp form of the Levi. These are called *Bessel periods* or *Fourier–Jacobi periods*, depending on the parity of the codimension of the Levi. For example, the q-expansion of a Siegel modular form is related to a Bessel period, whereas the Fourier–Jacobi expansion of a Siegel modular form, or in general other holomorphic automorphic forms on Shimura varieties, is related to, as the name suggests, a Fourier–Jacobi period.

On the other hand, the conjecture of Gan–Gross–Prasad (abbreviated GGP) suggests a relation between Bessel/Fourier–Jacobi periods with certain L-values. Although there are many instances of "refined" GGP conjectures not written in the literature, especially for the Fourieri–Jacobi case, the end result should be of the form

$$\frac{\text{Bessel/FJ period}}{\text{Petersson norm of holomorphic form}} \sim \frac{L(1,\mathrm{Ad})}{L(1/2,\pi_1)L(1/2,\pi_2)}.$$

Since the Bessel/Fourier–Jacobi period being rational can detect rationality, these two ways of detecting algebraicity should be related. As we will see in the two concrete instances of Sp_4 and $\mathrm{SU}(2,2)$, these two will be related via the framework of motivic periods and Beilinson conjectures.

5.2. **Lapid–Mao conjecture.** The Lapid–Mao conjecture [LM], on the other hand, is about the Petersson norm of *generic automorphic forms*. It has a universal formulation as follows.

Conjecture 5.1 ([LM]). *Let Π be a globally generic representation, satisfying (NEW) of Assumption 2.1. Let $f = \otimes_p f_p \in \Pi$ be a newform. Then,*

$$\frac{\langle f, f\rangle}{|W(1)|^2} \sim_{\mathbb{Q}^\times} \frac{L(1,\Pi,\mathrm{Ad})}{\Delta_G(1)|W_\infty(1)|^2},$$

where $\langle , \rangle$ is the L^2-norm (as opposed to the Petersson norm), $\Delta_G(s)$ is the L-function of the dual to the Artin motive attached to G as defined in [Gr]*, and W (W_∞, respectively) is the Whittaker function of f (f_∞, respectively).*

5.3. **The case of Sp_4.** We now put the relevant Hodge-linear algebra that proves the case of Sp_4. For the sake of simplicity, we assume we work with a parallel weight $(2,2)$ Siegel modular form, or Harish–Chandra paratemer $(1,0)$, although the Hodge-linear algebra calculation stays the same.

The motive M of Π should be of rank 4 and weight 1, with the Hodge decomposition

$$H_B(M)\otimes_{\mathbb{Q}}\mathbb{C} = \underbrace{\mathbb{C}f_{\mathrm{hol}} \oplus \mathbb{C}f_{\mathrm{gen}}}_{H^{1,0}(M)} \oplus \underbrace{\mathbb{C}\overline{f_{\mathrm{gen}}} \oplus \mathbb{C}\overline{f_{\mathrm{hol}}}}_{H^{0,1}(M)}.$$

Let e_1^+, e_2^+ be a $\mathbb{Q}$-basis of $H_B(M)^+$ and e_1^-, e_2^- be a $\mathbb{Q}$-basis of $H_B(M)^-$. Let $f_1, f_2 \in F^1H_{\mathrm{dR}}(M)$ be a $\mathbb{Q}$-basis, and $g_1, g_2 \in H_{\mathrm{dR}}(M)/F^1H_{\mathrm{dR}}(M)$ be a $\mathbb{Q}$-basis, and $\widetilde{g}_1, \widetilde{g}_2 \in H_{\mathrm{dR}}(M)$ be lifts of g_1, g_2. Given two $\mathbb{C}$-bases of $H_B(M)\otimes\mathbb{C} \cong H_{\mathrm{dR}}(M)\otimes\mathbb{C}$, we can write an expression

$$\begin{pmatrix} e_1^+ & e_2^+ & e_1^- & e_2^- \end{pmatrix} = \begin{pmatrix} f_1 & f_2 & \widetilde{g}_1 & \widetilde{g}_2 \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

for $A, B, C, D \in M_2(\mathbb{C})$. Note that by definition $\delta(M) = \det\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$.

We have canonical isomorphisms

$$H^{1,0}(M) \cong F^1H_{\mathrm{dR}}(M)\otimes\mathbb{C}, \quad H^{0,1}(M) \cong \frac{H_{\mathrm{dR}}(M)}{F^1H_{\mathrm{dR}}(M)}\otimes\mathbb{C}.$$

Let $f_{1,B}, f_{2,B} \in H^{1,0}(M)$ and $g_{1,B}, g_{2,B} \in H^{0,1}(M)$ be the images of f_1, f_2, g_1, g_2 under the above canonical isomorphisms. Then $f_{i,B}$ and f_i coincide as elements of $H_B(M)\otimes\mathbb{C} \cong H_{\mathrm{dR}}(M)\otimes\mathbb{C}$, whereas $\widetilde{g}_i - g_{i,B} \in H^{1,0}(M)$. So,

$$\begin{pmatrix} f_1 & f_2 & \widetilde{g}_1 & \widetilde{g}_2 \end{pmatrix} = \begin{pmatrix} f_{1,B} & f_{2,B} & g_{1,B} & g_{2,B} \end{pmatrix} \begin{pmatrix} 1_2 & M \\ 0_2 & 1_2 \end{pmatrix}.$$

In particular, if we write

$$\begin{pmatrix} e_1^+ & e_2^+ & e_1^- & e_2^- \end{pmatrix} = \begin{pmatrix} f_{1,B} & f_{2,B} & g_{1,B} & g_{2,B} \end{pmatrix} \begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix},$$

then $\det\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) = \det\left(\begin{smallmatrix} A' & B' \\ C' & D' \end{smallmatrix}\right)$.

Also, there must be relations

$$c_B(f_{1,B}) = ag_{1,B} + bg_{2,B},$$
$$c_B(f_{2,B}) = cg_{1,B} + dg_{2,B}.$$

Using that $F_\infty e_i^+ = e_i^+$, $F_\infty e_i^- = -e_i^-$ and that $F_\infty(f_{i,B}) = c_B(f_{i,B})$ and $F_\infty(g_{i,B}) = c_B(g_{i,B})$ (as in [PV, Lemma, §8.2.1]), one has

$$\begin{aligned} c_B(A'_{1i}f_{1,B} + A'_{2i}f_{2,B}) &= C'_{1i}g_{1,B} + C'_{2i}g_{2,B}, \\ c_B(B'_{1i}f_{1,B} + B'_{2i}f_{2,B}) &= -(D'_{1i}g_{1,B} + D'_{2i}g_{2,B}), \end{aligned}$$

for $i = 1, 2$. This can be packaged into $C' = \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) A'$, $D' = -\left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) B'$. So

$$\delta(M) = \det\begin{pmatrix} A' & B' \\ \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) A' & -\left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) B' \end{pmatrix} = \det\begin{pmatrix} A' & B' \\ 0_2 & -2\left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) B' \end{pmatrix} = 4\det\begin{pmatrix} a & c \\ b & d \end{pmatrix} \det A' \det B'.$$

Note on the other hand that $c^+(M) = \det C$, $c^-(M) = \det D$. As $C' = C$, $D' = D$, we see that

$$c^+(M)c^-(M) \sim_{\overline{\mathbb{Q}}^\times} \delta(M) \det\begin{pmatrix} a & c \\ b & d \end{pmatrix}.$$

Note also that $\operatorname{Ad} M = (\operatorname{Ad} M)^* = (\operatorname{Sym}^2 M)(1)$, which simplifies our calculation. We are also led to calculate $\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M)$ (we know it does not depend on the choice of weak polarization up to $\overline{\mathbb{Q}}^\times$-ambiguity). We know that $(\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M))^2 \sim_{\overline{\mathbb{Q}}^\times} \lambda$, where $\varphi(c_B(v^+)) = \lambda v^-$. Here, v^+, v^- are $\mathbb{Q}$-bases vectors for $\det F^1 H_{\mathrm{dR}}(\operatorname{Ad} M)$ and $\det\left(\frac{H_{\mathrm{dR}}(\operatorname{Ad} M)}{F^0 H_{\mathrm{dR}}(\operatorname{Ad} M)}\right)$, respectively, and $\varphi : \wedge^{\dim F^1 H_{\mathrm{dR}}(\operatorname{Ad} M)}(H_{\mathrm{dR}}(M) \otimes \mathbb{C}) \to \det\left(\frac{H_{\mathrm{dR}}(\operatorname{Ad} M)}{F^0 H_{\mathrm{dR}}(\operatorname{Ad} M)}\right)$ is the natural projection. As $\operatorname{Ad} M = (\operatorname{Sym}^2 M)(1)$, we can take f_1^2, f_1f_2, f_2^2 and g_1^2, g_1g_2, g_2^2 as $\mathbb{Q}$-bases of $F^1 H_{\mathrm{dR}}(\operatorname{Ad} M)$ and $\frac{H_{\mathrm{dR}}(\operatorname{Ad} M)}{F^0 H_{\mathrm{dR}}(\operatorname{Ad} M)}$, respectively. Now from the known relations,

$$\begin{aligned} c_B(f_1^2) &\equiv a^2g_1^2 + 2abg_1g_2 + b^2g_2^2, \\ c_B(f_1f_2) &\equiv acg_1^2 + (ad+bc)g_1g_2 + bdg_2^2, \\ c_B(g_1g_2) &\equiv c^2g_1^2 + 2cdg_1g_2 + d^2g_2^2, \end{aligned}$$

where $\equiv$ is mod $F^0 H_{\mathrm{dR}}(\operatorname{Ad} M) \otimes \mathbb{C}$. So

$$\lambda = \det\begin{pmatrix} a^2 & 2ab & b^2 \\ ac & ad+bc & bd \\ c^2 & 2cd & d^2 \end{pmatrix} = (ad-bc)^3.$$

Therefore, we obtain

$$\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M) \sim_{\overline{\mathbb{Q}}^\times} \frac{\sqrt{c^+(M)c^-(M)}^3}{\delta(M)^{3/2}}.$$

This is, by Deligne's conjectures, conjecturally the same as

$$\operatorname{vol} F^1 H_{\mathrm{dR}}(\operatorname{Ad} M) \sim_{\overline{\mathbb{Q}}^\times} \frac{\sqrt{L(1/2,\Pi)L(1/2,\Pi\otimes\chi_{\mathbb{Q}(i)})}^3}{\pi^3}.$$

Applying the period conjectures on this case, we obtain the following formulae.

Conjecture 5.2. *Let $\Pi = \Pi_f \otimes \Pi_{\mathrm{gen}}$ be as above. Let f_{hol}, f_{gen} be newforms (see Assumption 2.1) in $\Pi_f \otimes \Pi_{\mathrm{hol}}$, $\Pi_f \otimes \Pi_{\mathrm{gen}}$, respectively, where Π_{hol} is the corresponding holomorphic NLDS in the L-packet of Π_{gen}.*

(1) (Böcherer's conjecture; [Li], [FM]*)If B is the Bessel period of f_{hol} corresponding to the unitary character $\chi_{\mathbb{Q}(i)}$, then*

$$\frac{\langle f_{\mathrm{hol}}, f_{\mathrm{hol}}\rangle}{|B(1)|^2} \sim_{\overline{\mathbb{Q}}^\times} \pi^{-6} \frac{L(1,\Pi,\mathrm{Ad})}{L(1/2,\Pi)L(1/2,\Pi\otimes\chi_{\mathbb{Q}(i)})}.$$

(2) ([LM], [CI]*)*

$$\frac{\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle}{|W(1)|^2} \sim_{\overline{\mathbb{Q}}^\times} \pi^{9} \cdot \frac{L(1,\Pi,\mathrm{Ad})}{|W_\infty(1)|^2}.$$

Proof of Theorem 4.1 for Sp_4. We know that $B(1)$, being a $\overline{\mathbb{Q}}$-linear combination of Fourier coefficients of q-expansion of f_{hol}, being in $\overline{\mathbb{Q}}$ is equivalent to f_{hol} being defined over $\overline{\mathbb{Q}}$. For the algebraicity of f_{gen}, note that Corollary 4.1 is about the anti-generic LDS in our terminology, namely those appearing in H^2 of coherent cohomology. Thus, if f_{gen} generates an algebraic cohomology class over $\overline{\mathbb{Q}}$, then using that

$$\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P = (2\pi i)^3 \langle [f_{\mathrm{gen}}], [\overline{f}_{\mathrm{gen}}]\rangle_{\mathrm{coh}},$$

where $\langle -,-\rangle_{\mathrm{coh}}$ is the cohomological cup product defined over $\overline{\mathbb{Q}}$, we see that the RHS is $\sim_{\overline{\mathbb{Q}}^\times}$ $\pi^3 W_{\overline{f}_{\mathrm{gen}}}(1)\Lambda(1/2,\Pi)\Lambda(1/2,\Pi\otimes\chi_{\mathbb{Q}(i)})$. Since $W_{\overline{f}_{\mathrm{gen}}}(1) = \overline{W_{f_{\mathrm{gen}}}(1)}$, we have

$$\pi^9 \frac{L(1,\Pi,\mathrm{Ad})}{|W_\infty(1)|^2}|W_{f_{\mathrm{gen}}}(1)|^2 \sim_{\overline{\mathbb{Q}}^\times} \langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle \sim_{\overline{\mathbb{Q}}^\times} \pi^3 \overline{W_{f_{\mathrm{gen}}}(1)}\Lambda(\frac{1}{2},\Pi)\Lambda(\frac{1}{2},\Pi\otimes\chi_{\mathbb{Q}(i)}),$$

or

$$W(1) \sim_{\overline{\mathbb{Q}}^\times} \pi^{-6}\frac{|W_\infty(1)|^2\Lambda(\frac{1}{2},\Pi)\Lambda(\frac{1}{2},\Pi\otimes\chi_{\mathbb{Q}(i)})}{L(1,\Pi,\mathrm{Ad})} \sim_{\overline{\mathbb{Q}}^\times} \pi^{-10}\frac{|W_\infty(1)|^2 L(\frac{1}{2},\Pi)L(\frac{1}{2},\Pi\otimes\chi_{\mathbb{Q}(i)})}{L(1,\Pi,\mathrm{Ad})}.$$

We are to compute $\frac{\langle f_{\mathrm{hol}}, f_{\mathrm{hol}}\rangle_P}{\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P}$, which leads to

$$\begin{aligned}
\frac{\langle f_{\mathrm{hol}}, f_{\mathrm{hol}}\rangle_P}{\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P} &\sim_{\overline{\mathbb{Q}}^\times} \frac{\pi^{-6}\frac{L(1,\Pi,\mathrm{Ad})}{L(1/2,\Pi)L(1/2,\Pi\otimes\chi_{\mathbb{Q}(i)})}}{\pi^9|W(1)|^2\frac{L(1,\Pi,\mathrm{Ad})}{|W_\infty(1)|^2}} \\
&= \pi^{-15}\frac{|W_\infty(1)|^2}{|W(1)|^2 L(1/2,\Pi)L(1/2,\Pi\otimes\chi_{\mathbb{Q}(i)})} \\
&\sim_{\overline{\mathbb{Q}}^\times} \pi^{-15}\frac{|W_\infty(1)|^2}{\pi^{-20}\frac{|W_\infty(1)|^4 L(1/2,\Pi)^3 L(1/2,\Pi\otimes\chi_{\mathbb{Q}(i)})^3}{L(1,\Pi,\mathrm{Ad})^2}} \\
&\sim_{\overline{\mathbb{Q}}^\times} \frac{\pi^9}{|W_\infty(1)|^2}\left(\frac{L_\infty(1,\Pi,\mathrm{Ad})}{L_\infty(0,\Pi,\mathrm{Ad})}\cdot\frac{L(1,\Pi,\mathrm{Ad})}{\frac{\sqrt{L(1/2,\Pi)L(1/2,\Pi\otimes\chi_{\mathbb{Q}(i)})}^3}{\pi^3}}\right)^2,
\end{aligned}$$

which involves elementary calculation of archimedean L-factors. The parenthized term is precisely the RHS of Conjecture 3.2, which finishes the proof. □

Remark 5.1. Unfortunately, for now, the author has been unable to calculate $W_\infty(1)$. It is however very believable that the archimedean zeta integral against a preferred, nice test vector is equal to an archimedean local L-factor, which is $\sim_{\overline{\mathbb{Q}}^\times}$ half-integral powers of π. In our case, there is even an explicit integral expression of $W_\infty(1)$, as written in [CI]:

$$W_\infty(1) = 16e^{-2\pi}\pi^{\frac{7}{2}} \int_{c-i\infty}^{c+i\infty} \pi^{-2s}\Gamma(s+\frac{1}{2})^2 U(s+\frac{1}{2},1,4\pi)\Gamma(s)\frac{ds}{2\pi i},$$

where $U(a,b,z) = \frac{1}{\Gamma(a)}\int_0^\infty e^{-zt}t^{a-1}(1+t)^{b-a-1}dt$ is the confluent hypergeometric function of the second kind.

5.4. **The case of** $\mathrm{SU}(2,1)$**.** This case, we exploit the fact that H^1 is the middle degree of the Shimura variety. Let f_{gen} be a newform in a globally generic Π satisfying Assumption 2.1 and that Π_∞ is a generic NLDS whose infinitesimal character lies on the wall between the holomorphic chamber and the generic chamber. Then, $\overline{f}_{\mathrm{gen}}$ is in the generic LDS as well (with different infinitesimal character), and for both we can use Corollary 4.2 to detect rationality. Suppose further that f_{gen} gives rise to a cohomology class $[f_{\mathrm{gen}}]$ in H^1 defined over $\overline{\mathbb{Q}}$. We have

$$\begin{aligned}\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P &= (2\pi i)^2\langle [f_{\mathrm{gen}}], [\overline{f}_{\mathrm{gen}}]\rangle_{\mathrm{coh}} \\ &\sim_{\overline{\mathbb{Q}}^\times} \pi^2 W_{\overline{f}_{\mathrm{gen}}}(1)\Lambda(1/2, BC(\Pi)).\end{aligned}$$

On the other hand, the Lapid–Mao conjecture says

$$\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P \sim_{\mathbb{Q}^\times} |W_{f_{\mathrm{gen}}}(1)|^2 \frac{L(1,\Pi,\mathrm{Ad})}{|W_\infty(1)|^2},$$

so by using $W_{\overline{f}_{\mathrm{gen}}}(1) = \overline{W_{f_{\mathrm{gen}}}(1)}$,

$$W_{f_{\mathrm{gen}}}(1) \sim_{\overline{\mathbb{Q}}^\times} \pi^2 \frac{|W_\infty(1)|^2\Lambda(1/2, BC(\Pi))}{L(1,\Pi,\mathrm{Ad})}.$$

On the other hand, Corollary 4.2 says

$$W_{f_{\mathrm{gen}}}(1) \sim_{\overline{\mathbb{Q}}^\times} \frac{1}{\Lambda(1/2, BC(\Pi))},$$

which gives an extra relationship that we can utilize; namely,

$$\Lambda(1,\Pi,\mathrm{Ad}) \sim_{\overline{\mathbb{Q}}^\times} \Lambda(1/2, BC(\Pi))^2|W_\infty(1)|^2\pi^2.$$

Note that this is already observed in [Ha4, Corollary 1.3.5], under the assumption of Deligne's conjectures. In particular, assuming Deligne's conjectures, we deduce that $|W_\infty(1)| \sim_{\overline{\mathbb{Q}}^\times} \pi^{blah}$.

On the other hand, the refined Gan–Gross–Prasad conjecture (in this case it is often referred also as Ichino–Ikeda conjecture; cf. [PV, §6.7]) says

$$\langle f_{\mathrm{hol}}, f_{\mathrm{hol}}\rangle_P \sim_{\overline{\mathbb{Q}}^\times} \pi^{-2}\frac{L(1,\Pi,\mathrm{Ad})}{L(1/2, BC(\Pi))}.$$

Combining these, we get the desired result as follows.

Proof of Theorem 4.1 for $\mathrm{SU}(2,1)$. For calculational simplicity, we assume that we work with the case of Harish-Chandra character $(1,1,0)$. We know that

$$\begin{aligned}\frac{\langle f_{\mathrm{hol}}, f_{\mathrm{hol}}\rangle_P}{\langle f_{\mathrm{gen}}, f_{\mathrm{gen}}\rangle_P} &\sim_{\overline{\mathbb{Q}}^\times} \frac{\pi^{-2}\frac{L(1,\Pi,\mathrm{Ad})}{L(1/2,BC(\Pi))}}{\pi^4\frac{|W_\infty(1)|^2\Lambda(1/2,BC(\Pi))^2}{L(1,\Pi,\mathrm{Ad})}} \\ &\sim_{\overline{\mathbb{Q}}^\times} \frac{1}{|W_\infty(1)|^2}\frac{L(1,\Pi,\mathrm{Ad})^2}{L(1/2,BC(\Pi))^3} \\ &\sim_{\overline{\mathbb{Q}}^\times} \frac{\pi^{18}}{|W_\infty(1)|^2}\left(\frac{L_\infty(1,\Pi,\mathrm{Ad})}{L_\infty(0,\Pi,\mathrm{Ad})}\cdot\frac{L(1,\Pi,\mathrm{Ad})}{L(1/2,BC(\Pi))^{3/2}}\right)^2,\end{aligned}$$

so we would like to show that $\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\,\Pi) \sim_{\overline{\mathbb{Q}}^\times} L(1/2, BC(\Pi))^{3/2}$ (times a power of π). In this case, the motive M associated with Π is a motive over F with coefficients in $\mathbb{Q}$, where if we denote $\sigma : F \to \mathbb{C}$ by the preferred complex embedding, then as the minimal K-type is just $(1,1,1)$, by the recipe in [Ha4, 1.1.10],

$$H_B(M_\sigma)\otimes_{\mathbb{Q}}\mathbb{C} = \underbrace{\mathbb{C}f_1\oplus\mathbb{C}f_2\oplus\mathbb{C}f_3}_{H^{1,0}(M)}, \quad H_B(M_{\overline{\sigma}})\otimes_{\mathbb{Q}}\mathbb{C} = \underbrace{\mathbb{C}\overline{f}_1\oplus\mathbb{C}\overline{f}_2\oplus\mathbb{C}\overline{f}_3}_{H^{0,1}(M)},$$

which is, in terms of Hodge-linear algebra, the same as the direct sum of three copies of unitary Hecke character of F with infinity type $(1,0)$. In that case, it is easily seen that $(\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\, M))^2 \sim_{\overline{\mathbb{Q}}^\times} L(1/2, M)$, by Stark's conjecture for imaginary quadratic fields (cf. [Ta, §IV]). Thus, our motive satisfies $\mathrm{vol}\, F^1 H_{\mathrm{dR}}(\mathrm{Ad}\,\Pi) \sim_{\overline{\mathbb{Q}}^\times} L(1/2, BC(\Pi))^{3/2}$, as desired. □

References

[AS] M. Atiyah, W. Schmid, A geometric construction of the discrete series for semisimple Lie groups. *Invent. Math.* 42 (1977), 1-62.

[Bö] S. Böcherer, Über die Fourier–Jacobi-Entwicklung Siegelscher Eisensteinreihen II. *Math. Z.* 189 (1985), 181-188.

[BHR] D. Blasius, M. Harris, D. Ramakrishnan, Coherent cohomology, limits of discrete series, and Galois conjugation. *Duke Math. J.* 73 (1994), no. 3, 647-685.

[BW] A. Borel, N. Wallach, Continuous cohomology, discrete subgroups, and representations of reductive groups. Second edition. Mathematical Survey and Monographs, 67. *American Mathematical Society, Providence, RI*, 2000.

[Ca] H. Carayol, Quelques relations entre les cohomologies des variétés de Shimura et celles de Griffiths–Schmid (cas du groupe $\mathrm{SU}(2,1)$). *Compositio Math.* 121 (2000), no. 3, 305-335.

[Cl] L. Clozel, Motifs et formes automorphes: applications du principe de fonctorialité. *Automorphic forms, Shimura varieties, and L-functions, Vol. I (Ann Arbor, MI, 1988)*, 77-159, Perspect. Math., 10, *Academic Press, Boston, MA*, 1990.

[CI] S.-Y. Chen, A. Ichino, On Petersson norms of generic cusp forms and special values of adjoint L-functions for GSp_4. Preprint, 2019. `https://arxiv.org/pdf/1902.06429v1.pdf`.

[DPSS] M. Dickson, A. Pitale, A. Saha, R. Schmidt, Explicit refinedments of Böcherer's conjecture for Siegel modular forms of squarefree level. *J. Math. Soc. Japan* 72 (2020), no. 1, 251-301.

[FM] M. Furusawa, K. Morimoto, Refined global Gross–Prasad conjecture on special Bessel periods and Böcherer's conjecture. *J. Eur. Math. Soc. (JEMS)* 23 (2021), no. 4, 1295-1331.

[GPS] S. Gelbart, I. Piatetski-Shapiro, Automorphic forms and L-functions for the unitary group. *Lie group representations, II (College Park, Md., 1982/1983)*, 141-184, Lecture Notes in Math., 1041, *Springer, Berlin*, 1984.

[GPSR] S. Gelbart, I. Piatetski-Shapiro, S. Rallis, Explicit constructions of automorphic L-functions. Lecture Notes in Mathematics, 1254. *Springer-Verlag, Berlin*, 1987.

[Go] Y. Gon, Generalized Whittaker functions on $\mathrm{SU}(2,2)$ with respect to the Siegel parabolic subgroup. *Mem. Amer. Math. Soc.* 155 (2002), no. 738.

[Gr] B. Gross, On the motive of a reductive group. *Invent. Math.* 130 (1997), no. 2, 287-313.

[GGK1] M. Green, P. Griffiths, M. Kerr, Hodge theory, complex geometry, and representation theory. CBMS Regional Conference Series in Mathematics, 118. *Published for the Conference Board of the Mathematical Sciences, Washington, DC; by the American Mathematical Society, Providence, RI*, 2013.

[GGK2] M. Green, P. Griffiths, M. Kerr, Special values of automorphic cohomology classes. *Mem. Amer. Math. Soc.* 231 (2014), no. 1088.

[GT] M. Goresky, Y. S. Tai, Anti-holomorphic multiplication and a real algebraic modular variety. *J. Differential Geom.* 65 (2003), no. 3, 513-560.

[Ha1] M. Harris, Automorphic forms of $\overline{\partial}$-cohomology type as coherent cohomology classes. *J. Differential Geom.* 32 (1990), no.21, 1-63.

[Ha2] M. Harris, Period invariants of Hilbert modular forms. I. Trilinear differential operators and L-functions. *Cohomology of arithmetic groups and automorphic forms (Luminy-Marseille, 1989)*, 155-202, Lecture Notes in Math., 1447, *Springer, Berlin*, 1990.

[Ha3] M. Harris, L-functions and periods of polarized regular motives. *J. Reine Angew. Math.* 483 (1997), 75-161.

[Ha4] M. Harris, L-functions and periods of adjoint motives. *Algebra Number Theory* 7 (2013), no. 1, 117-155.

[Ho] A. Horawa, Motivic action on coherent cohomology of Hilbert modular varieties. Preprint, 2020. `https://arxiv.org/pdf/2009.14400v1.pdf`.

[HV] M. Harris, A. Venkatesh, Derived Hecke algebra for weight one forms. *Exp. Math.* 28 (2019), no. 3, 342-361.

[Ja] F. Januszewski, Rational structures on automorphic representations. *Math. Ann.* 370 (2018), no. 3-4, 1805-1881.

[Ke] M. Kerr, Cup products in automorphic cohomology: the case of Sp_4. *Hodge theory, complex geometry, and representation theory*, 199-233, Contemp. Math., 608, *Amer. Math. Soc., Providence, RI*, 2014.

[Kn] A. Knapp, Representation theory of semisimple groups. An overview based on examples. Princeton Mathematical Series, 36. *Princeton University Press, Princeton, NJ*, 1986.

[Ko] B. Kostant, Lie algebra cohomology and the generalized Borel–Weil theorem. *Ann. of Math.* 74 (1961), 329-387.

[KO] H. Koseki, T. Oda, Whittaker functions for the large discrete series representations of $\mathrm{SU}(2,1)$ and related zeta integrals. *Publ. Res. Inst. Math. Sci.* 31 (1995), no. 6, 959-999.

[KZ1] A. Knapp, G. Zuckerman, Normalizing factors, tempered representations,a nd L-groups. *Automorphic forms, representations and L-functions (Proc. Sympos. Pure Math., Oregon State Univ., Corvallis, Ore., 1977), Part 1*, pp. 93-105, Proc. Sympos. Pure Math., XXXIII, *Amer. Math. Soc., Providence, R.I.*, 1979.

[KZ2] A. Knapp, G. Zuckerman, Classification of irreducible tempered representations of semisimple groups. *Ann. of Math.* 116 (1982), no. 2, 389-455.

[KZ3] A. Knapp, G. Zuckerman, Classification of irreducible tempered representations of semisimple groups. II. *Ann. of Math.* 116 (1982), no. 3, 457-501.

[La] R. Langlands, Notes on the Knapp–Zuckerman theory. Preprint, 1977. `http://publications.ias.edu/sites/default/files/notes-kz-theory-rpl.pdf`.

[Lan] K.-W. Lan, Closed immersions of toroidal compactifications of Shimura varieties. To appear in *Math. Res. Lett.*, 2021.

[Li] Y. Liu, Refined global Gan–Gross–Prasad conjecture for Bessel periods. *J. Reine Angew. Math.* 717 (2016), 133-194.

[LM] E. Lapid, Z. Mao, A conjecture on Whittaker–Fourier coefficients of cusp forms, *J. Number Theory* 146 (2015), 448-505.

[LPSZ] D. Loeffler, V. Pilloni, C. Skinner, S. Zerbes, Higher Hida theory and p-adic L-functions for GSp_4. To appear in *Duke Math. J.*, 2021.

[Mu] G. Muić, Intertwining operators and composition series of generalized and degenerate principal series for $\mathrm{Sp}(4,\mathbb{R})$. *Glas. Mat. Ser. III* 44(64) (2009), no. 2, 349-399.

[No] M. Novodvorsky, Automorphic L-functions for the symplectic group $\mathrm{GSp}(4)$. *Automorphic forms, representations and L-functions (Proc. Sympos. Pure Math., Oregon State Univ., Corvallis, Ore., 1977), Part 2*, 87-95, Proc. Sympos. Pure Math., XXXIII, *Amer. Math. Soc., Providence, R.I.*, 1979.

[Od1] T. Oda, An explicit integral representation of Whittaker functions on $\mathrm{Sp}(2;\mathbb{R})$ for the large discrete series representations. *Tohoku Math. J.* 46 (1994), no. 2, 261-279.

[Od2] T. Oda, Real harmonic analysis for geometric automorphic forms. Preprint, 2005. `https://www.ms.u-tokyo.ac.jp/~takayuki/Preprints/oda-texel.pdf`.

[Oh] G. Oh, Higher Hida theory and the p-adic L-functions for non-holomorphic automorphic representations. Preprint, 2021.

[Po] A. Pollack, Rankin–Selberg integrals in many complex variables and Rankin–Selberg integrals associated to non-unique models. Thesis (Ph.D.)-Princeton University. 2014.

[Pr] D. Prasad, Theta correspondence for unitary groups. *Pacific J. Math.* 194 (2000), no. 2, 427-438.

[PV] K. Prasanna, A. Venkatesh, Automorphic cohomology, motivic cohomology, and the adjoint L-function. To appear in *Astérisque*, 2021.

[Ro] J. Rogawski, Automorphic representations of unitary groups in three variables. Annals of Mathematics Studies, 123. *Princeton University Press, Princeton, NJ*, 1990.

[Sc1] W. Schmid, On a conjecture of Langlands. *Ann. of Math.* 93 (1971), 1-42.

[Sc2] W. Schmid, On the characters of the discrete series. The Hermitian symmetric case. *Invent. Math.* 30 (1975), no. 1, 47-144.

[Sh1] G. Shimura, On the periods of modular forms. *Math. Ann.* 229 (1977), no. 3, 211-221.

[Sh2] G. Shimura, The special values of the zeta functions associated with Hilbert modular forms. *Duke Math. J.* 45 (1978), no. 3, 637-679.

[Sh3] G. Shimura, Arithmeticity in the theory of automorphic forms. Mathematical Surveys and Monographs, 82. *American Matehmatical Society, Providence, RI*, 2000.

[Su] J. Su, Coherent cohomology of Shimura varieties and automorphic forms. Preprint, 2018. `https://arxiv.org/pdf/1810.12056v1.pdf`.

[Ta] J. Tate, Les conjectures de Stark des sur les fonctions L d'Artin en $s = 0$. Lecture notes edited by Dominique Bernardi and Norbert Schappacher. Progress in Mathematics, 47. *Birkhäuser Boston, Inc., Boston, MA*, 1984.

[Va] Y. Varshavsky, On the characterization of complex Shimura varieties. *Selecta Math. (N.S.)* 8 (2002), no. 2, 283-314.

[Vo1] D. Vogan, Representations of real reductive groups. Progress in Mathematics, 15. *Birkhäuser, Boston, Mass.*, 1981.

[Vo2] D. Vogan, The local Langlands conjecture. *Representaiton theory of groups and algebras*, 305-379, Contemp. Math., 145, *Amer. Math. Soc., Providence, RI*, 1993.

[VZ] D. Vogan, G. Zuckerman, Unitary representations with nonzero cohomology. *Composition Math.* 53 (1984), no. 1, 51-90.

[Wa] N. Wallach, On the Selberg trace formula in the case of compact quotient. *Bull. Amer. Math. Soc.* 82 (1976), no. 2, 171-195.

[Wi] F. Williams, The n-cohomology of limits of discrete series. *J. Funct. Anal.* 80 (1988), no. 2, 451-461.

[Yo] H. Yoshida, Motives and Siegel modular forms. *Amer. J. Math.* 123 (2001), no. 6, 1171-1197.